

University of Sheffield

Visualising Emotion: A Similarity-Based Analysis of Emotional Fidelity in Text-to-Image Generation

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Declaration

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Abstract

At the Festival of the Mind in Sheffield in 2024, visitors typed a few words about how they were feeling and a text-to-image model returned a monochrome line drawing of each entry, producing 1,025 prompt–drawing records of which 1,012 are analysed here. This dissertation asks whether the emotion recoverable from the drawing agrees with the emotion recoverable from the words. There is no ground truth, so two automatic readers stand in for the two sides: a fine-tuned RoBERTa model reads the prompt, and a describe-then-extract reader, in which a visual question-answering model answers five factual questions about the drawing and a natural-language-inference model infers an emotion from the resulting sentence, reads the image. Agreement is measured as similarity rather than exact match, on a ladder of four measures from valence–arousal distance to top-1 family, and every value is read against a random-pairing null, because the loosest measure has a floor of 0.41 and randomly paired records already score 76.6% on it. Intervals and tests cluster on prompt groups, since 249 records repeat a prompt.

On the 607 prompts the text reader labels emotional, valence-sign agreement is 64.0% against a null of 45.6% ($z = +11.1$) and family agreement 53.0% against 38.8%; the results survive a distinct-prompt re-run and a Holm correction. Drawings in which the describer reads a face agree far more than drawings of scenes, 73.9% against 42.9%, a difference of +30.9 pp (95% CI [+22.6, +40.6], clustered permutation $p = 0.0001$). Prompts the reader labels neutral sit at chance. Scored like for like, describe-then-extract agrees with the prompt reader on 53.0% of families against 22.4% for a direct CLIP reader. When the same words are drawn again, the image reader returns the same family for 68.8% of drawings in a group against 59.4% by chance.

The audits qualify these results. The describer names a facial expression for 406 of the 414 scene drawings in which it reports no person; where its expression word names an emotion, the image reader copies it 99.8% of the time; and a template ablation shows that removing that one word removes nearly all of the agreement, while a people-aware description leaves the face results unchanged and brings the scene results down to chance. The findings are therefore about a measurement chain rather than about human perception, and the dissertation ends with the single-coder check and the human-rated study that would convert reader agreement into perceived affect. The contribution is the corpus and question, the null-referenced similarity ladder, the consistency experiment the repeated prompts make possible, and a reproducible, audited toolkit with a live demonstrator.

Glossary

Record One prompt–drawing pair from the corpus. Records are not linked to individuals; 1,025 were processed and 1,012 analysed.

Prompt reader The text model that reads the visitor's words: GoEmotions (RoBERTa fine-tuned on 28 labels), collapsed to seven families.

Describer The vision model that answers five factual questions about a drawing (BLIP-VQA). It is never asked for an emotion.

Image reader The text model that infers an emotion family from the describer's one-sentence description (BART-MNLI, zero-shot).

Family One of seven emotion families: joy, sadness, anger, fear, surprise, disgust, neutral (Ekman's six plus neutral).

Agreement The central quantity: how similar the prompt reader's reading of the words is to the image reader's reading of the drawing. It is a property of two machine readers, not a statement about what a person would see.

Valence sign + for joy, − for sadness, anger, fear and disgust, 0 for surprise and neutral. Valence-sign agreement is computed over records whose prompt family has a sign.

Family agreement The share of records whose top-1 families coincide.

Top-3 soft overlap The overlap of the two readers' top-3 distributions after each is normalised to sum to one.

Valence–arousal similarity One minus the normalised distance between the two readings' score-weighted centroids in valence–arousal space. Its floor over the family centroids is 0.412, and randomly paired records already average about 76.6%.

Random-pairing null The mean and spread of a measure when the image readings are re-paired with the prompts at random (300 re-pairings, seed 7). Every reported value is read against it.

z (observed − null mean) / null standard deviation. An ordering statistic here, not a precise probability: across seeds it moves by about one unit (+11.2 to +12.2 for the headline).

Degenerate null A null with zero spread, which arises when every prompt in a subset shares one label so permutation cannot change the score. No z is reported for it.

Prompt group / dedup_key All records whose prompt text (before redaction, case-folded) is identical. 763 groups; 249 records repeat an earlier prompt. Every resampling and permutation clusters on it.

Clustered bootstrap Resampling prompt groups, not records, with replacement (10,000 draws) to obtain 95% confidence intervals that respect repeated prompts.

Face-classified / scene-classified A record is face-classified when the describer reports at least one person *and* its expression answer contains a readable expression word; every other record is scene-classified. The split is the describer's, not a human's.

Emotional / neutral-labelled prompt A prompt is emotional when the prompt reader's top-1 family is not neutral (607); otherwise it is neutral-labelled (405). The label is the reader's judgement, never human-checked.

Specificity check The expectation that neutral-labelled prompts show no agreement above their null. Only the valence–arousal measure is informative there.

Copy rate Where the describer's expression answer names an emotion word, the share of records in which the image reader's top-1 family equals that word's family (99.8% on the strict definition).

Consistency For prompts drawn more than once, the mean share of drawings in a group that the image reader assigns the group's modal family, read against a permutation null.

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Chapter 1

Introduction

1.1 Context

In September 2024 the University of Sheffield ran an installation at the Festival of the Mind in which members of the public were invited to type a few words about how they were feeling. Each entry was sent to a text-to-image model, which returned a black-and-white line drawing, and the drawing was shown back to the visitor. Over eleven festival days the installation produced 1,025 prompt–drawing records. That corpus, handed to me with no labels, no metadata and no record of who typed what, is the object of this dissertation.

Two properties of the corpus shaped everything that follows. The first is that the entries are short and personal. The median prompt is four words long, a quarter of them are a single word, and the commonest entries are the plain words people use for their own state: *happy*, *tired*, *excited*. Many combine feelings that a single label cannot hold, such as "tired but happy" or "hopeful, slightly anxious, resigned". The second is that the drawings are monochrome. The installation appended a line-art style instruction to every prompt, so colour, which the literature identifies as the strongest carrier of emotional meaning in generated images [2], is absent from every one of the 1,025 images. Whatever feeling a drawing shows, it has to show it through line, posture, faces and composition alone.

The corpus also has a shape that has to be stated plainly at the outset. The interface accepted free text, and a substantial minority of entries are not self-report at all: requests for objects ("pigs"), pop culture, cars, questions addressed to the machine. Records are not linked to individuals. A visitor could submit several times, and 249 of the 1,012 analysed records repeat a prompt that appears earlier in the corpus, so the number of contributors is unknown and can be no larger than 763. These are properties of the collection, not defects in the analysis, and Chapter 3 explains how each is handled. Figure 1.1 shows six records as they appear to the reader of this dissertation: the visitor's words, redacted of any personal name, and the drawing that came back.



Figure 1.1: Six records from the corpus. The caption under each drawing gives the record number, the visitor's words, the image class assigned by the describer (Section 3.4), and the emotion family each reader returned. Records 63 and 201 are the two used in the live demonstration; record 21 is a mountain scene for which the describer nevertheless reported a smiling expression, a behaviour examined in Section 4.8.

1.2 The question

When a text-to-image model draws what a person wrote about their feeling, does the emotion recoverable from the drawing agree with the emotion recoverable from the words? That is the question, and it is deliberately narrower than it first sounds. There is no ground truth here. Nobody rated the drawings, and the visitor's own words are the only record of what they meant. Two readers therefore stand in for the two sides of the comparison: a text model that reads an emotion from the prompt, and a second reader that reads an emotion from the drawing. What the project measures is the agreement between those two readings, and nothing in it is a statement about what a person would see. The human study that would establish that is the natural next step, and the pipeline built here was designed to precede it.

The intellectual centre of the project is a reframing my supervisor proposed in July, and which the results bear out: similarity, not matching. The obvious way to score an image against a text label is right or wrong, and the obvious way to summarise such scores is an accuracy. Neither survives contact with emotion. People disagree about what a face shows; models are noisy; and the labels themselves overlap and blend. Cowen and Keltner [19] found that self-reported experience resolves into more than two dozen categories joined by continuous gradients, and Russell's circumplex [74, 67] places every one of them on a plane where "happy" and "excited" sit close together. Treating a drawing read as excited as a total failure for a prompt that said happy throws away exactly the information the corpus is rich in. The right instrument is a ladder of measures of increasing strictness, each read against what random pairing of drawings and words would give. The four rungs, defined in Chapter 3, are valence–arousal similarity, valence-sign agreement, emotion-family agreement and top-3 overlap. The random-pairing null turns out to matter as much as the ladder: the loosest measure cannot fall below 0.41 and randomly paired records already score about 76.6% on it, so a bare percentage reads as a pass when it may be below chance.

1.3 Aims and objectives

The aim is to measure, on this corpus and with every value referenced to its own chance level, how far the emotion two automatic readers recover from a visitor's words and from the drawing made of them agree, and to characterise the instrument that does the measuring well enough that its limits are known. The objectives were:

1. To build an image reader that recovers an emotion from a monochrome drawing without asking a vision model for one: the vision model is asked factual questions, and a separate text model infers the emotion from the answers.
2. To define a similarity ladder and a random-pairing null so that every reported value carries its chance level.
3. To measure agreement over the whole corpus and over the two splits my supervisor asked for: prompts the text reader labels emotional against those it labels neutral, and

drawings in which a face can be read against drawings of scenes.

4. To test the face/scene difference formally and to check that the results survive the repeated prompts in the corpus.
5. To compare the describe-then-extract reader with a direct CLIP reader like for like, on the same records and against the same prompt reader.
6. To use the repeated prompts to measure how stable the whole prompt-to-reading chain is when the same words are drawn again.
7. To audit the instrument: the control set, the describer, an error taxonomy of the strongest disagreements, an ablation of the description template, and a single-coder reliability check.
8. To release a reproducible toolkit with pinned numeric tests and a live demonstrator.

1.4 Contributions

I want to be careful about what is new here, because the toolkit is not. The models are existing ones, recombined: a fine-tuned RoBERTa on the prompt side, a visual question-answering model and a natural-language-inference model on the image side, and the agreement statistics are standard. What is new is the object, the question and the way the question is asked.

The object is a corpus of unmediated, mixed-affect public self-report rendered as colour-free line art. The closest prior study, by Lomas and colleagues [49], rated emotional alignment on a fixed ten-item typology, each emotion rendered on a person or a robot, and it did so in colour. Nothing comparable exists for real, short, blended entries with colour removed, and the review in Chapter 2 was unable to find one within a 73-paper corpus.

The question is end-to-end affect agreement on non-templated input, and the way it is asked is the null-referenced similarity ladder: four measures, each with a measured chance level, clustered intervals that respect the repeated prompts, and a formal test of the one contrast (faces against scenes) the design was built around. The repeated prompts also make a consistency experiment possible that neither the rating studies nor the affective-generation literature perform, because none of them draws the same words more than once.

The audits are a contribution of a different kind. They show, with numbers rather than a caveat, where the instrument is weak: how much of the image reading rests on a single word the describer produces, what the describer does when there is no face to describe, and which of the strongest disagreements are the generator's fault and which are the readers'. The dissertation ends less certain about scene drawings than it began, and I take that to be the right direction for a measurement study to move in.

The deliverables are the analysis code, the 1,012-record result tables, a live demonstrator that runs the real pipeline on any record or an uploaded image against the same chance levels, a two-layer test suite in which every headline number is pinned to the file that produces it, and the audits reported in Chapter 4. Every statistic in this document is a macro generated from those result files, so the text cannot drift from the analysis.

1.5 Overview of the report

Chapter 2 reviews the literature in eight themes, from emotion theory to the perception of AI-generated images, and states the gap. Chapter 3 describes the corpus, the pre-processing, the pipeline, the four measures, the null, the inference procedures, the analyses and the demonstrator, together with the alternatives that were tried and rejected. Chapter 4 reports the results, tables and figures first, with interpretation held back. Chapter 5 argues what the results establish, why the describer is the weak link, what is not claimed, and how the findings sit beside other people's work. Chapter 6 concludes and sets out further work that follows from the evaluation rather than from a wish list. The appendices give the repository layout and run order, and the instruments: the five questions, the template, the hypothesis, the label map, the expression-word list, the redaction procedure and the coding sheet.

Chapter 2

Literature Review

2.1 Scope and method of the review

The question sits between several literatures that rarely cite one another: emotion psychology, emotion recognition from text, text-to-image generation, the evaluation of generated images, and the perception of emotion in pictures. I searched each of them in turn, following citations outward from the eight papers my supervisor supplied at the start of the project, and read 97 papers in full. Each was scored on two axes, relevance to the dissertation's path and strength of evidence, and retained only if it was both on the path and empirical: a study with an evaluation, or a dataset, benchmark or lexicon that other studies use. Surveys, position pieces and work that was rigorous but off target were excluded, with the reason logged; 24 papers went that way. A handful of seminal foundations that would fail the evidence test on their own terms, Ekman's argument for basic emotions and Russell's circumplex among them, were kept by exception because everything downstream assumes them. The 73 retained papers fall into eight themes, and those themes structure this chapter. Each section ends by saying what the theme means for this project, because the point of the review is not coverage but the decisions it forced.

One note on words. The literature's own term for the correspondence between a prompt and its image is *alignment*, and I use that word when describing other people's work. When I speak about this project I use *agreement*, because what is measured here is the agreement of two readers, and the difference is not cosmetic.

2.2 Foundations: emotion theory and taxonomy

Two traditions supply the vocabulary. Ekman's argument for a small set of basic emotions with distinct signals [23] gives the discrete scheme that most classifiers inherit: happiness, sadness, anger, fear, surprise, disgust. Russell's circumplex [74], elaborated by Posner, Russell and Peterson [67], places affect on two continuous axes, valence and arousal, and argues that the neural and developmental evidence fits a dimensional account better than a categorical one. The two are not rivals for this project so much as two rungs of the same ladder: the

seven families used here (Ekman's six plus neutral) are the categorical rung, and each family's position on the circumplex supplies the dimensional one.

The evidence that neither scheme is the whole story comes from Cowen and Keltner [19], who collected self-reports to more than two thousand short video clips and found that experience resolves into 27 distinct categories connected by continuous gradients, with the categories explaining more variance than the dimensions but neither sufficient alone. That is the finding behind the similarity framing. If "excited" and "happy" lie on a gradient rather than in separate boxes, an instrument that scores them as a miss is measuring the scheme rather than the phenomenon.

The supervisor's own line of work sharpens the point for faces in particular. Jack and colleagues showed with reverse correlation that the six basic expressions are not represented the same way by Western and East Asian observers [36], and that dynamic expressions transmit signals in a hierarchy over time, with early cues collapsing six emotions into four and fear and surprise, disgust and anger confusable at the start [35]. Chen and colleagues found that pain and pleasure have distinct facial signatures within each culture [12], transferred psychologically valid expressions to social robots and virtual agents by measuring what people actually perceive rather than what a theory says they should [14, 13], and most recently showed that facial movements encode emotion category and intensity as separable components [15]. Three things follow for a project that reads emotion from drawn faces. There is no canonical drawing of an emotion; near misses between neighbouring categories are expected, not anomalous; and any claim about what a face signals rests, in the end, on human perception, which this project does not measure.

For this project the foundations settle two design choices: a family scheme small enough for two very different readers to share, and a dimensional measure alongside it so that a near miss counts for something.

2.3 Emotion from text

Emotion recognition from text is mature on the material it was built for, and demonstrably weakest on the material this corpus contains. GoEmotions [21] is the reference resource: 58,000 Reddit comments annotated with 27 fine-grained categories plus neutral, with a grouping of those categories onto Ekman's families proposed by the authors. Its BERT baseline reached a macro-F1 of 0.46, and it did so unevenly, collapsing towards zero on rare, implicit categories. Harutyunyan and Kumar's unified benchmark on the same data [28] confirms the shape of that result across classical, recurrent and transformer models. Yang and colleagues [84] audited emotion datasets with large language models and found that very short texts are systematically under-specified: the label often depends on context the text does not contain. The register of the present corpus, first-person, a few words long, and frequently mixed, is precisely that case. Liu and colleagues [48] address short texts directly with a multi-label method, and Bhat [5] reports that short English texts can be classified well with a small transformer, but both work on tweets, which carry more context than "tired

but happy".

Dimensional resources offer a partial way round the problem. EmoBank [8] annotates ten thousand sentences with valence, arousal and dominance from both the writer's and the reader's perspective, and shows the two do not coincide, which matters here because a visitor's words are writer-perspective and every reader in this project is, by construction, a reader. The NRC lexicons give word-level valence, arousal and dominance norms for twenty thousand [56] and now fifty-five thousand [57] English words, and the earlier crowdsourced word–emotion lexicon [60] maps words onto basic emotions. LIWC [77] remains the transparent baseline for psychologists: a validated dictionary of positive, negative, anxious, angry and sad words. Zanwar and colleagues [89] show that fusing psycholinguistic features with a transformer improves out-of-domain generalisation, which is a live concern for a corpus unlike any training set. Niu and Mower Provost [62] distil emotion knowledge from a large language model into a compact zero-shot recogniser, and Parkington and colleagues [65] compare a large language model against expert human coders on thematic analysis and report both high agreement and a loss of nuance on fine-grained emotional triggers. Topic modelling of the BERTopic kind [27] was considered for surfacing themes across the corpus and set aside: with a median of four words the documents are too sparse for it.

The method references for the text side are the models themselves. Zero-shot classification by entailment [88] turns a label into a hypothesis and asks a natural-language-inference model whether the text entails it; BART [42] fine-tuned on the MultiNLI corpus [79] is the standard such model, and it is the image-side reader here. The text-to-text framing of T5 [69] and its instruction-tuned descendant [17] underlie several of the vision–language models discussed below. All of them are run through the Transformers library [81].

For this project the lesson is uncomfortable and has to be built into the design: the prompt reader is the weakest link on the text side, its labels on four-word entries are judgements rather than facts, and every downstream number inherits that uncertainty. The choice of GoEmotions over the alternatives is defended in Chapter 3; the decision to treat its neutral label as a group to be analysed rather than a filter to be applied is a direct consequence of this literature.

2.4 Datasets, benchmarks and lexicons

The image-side resources are, without exception, photographic or painted and coloured. EmoSet [86] is the largest, 118,000 images labelled with eight emotions and rich attributes; the affective-image classification of Machajdik and Hanbury [50] built features from colour, texture and composition and reported a human-agreement baseline; the Nencki Affective Picture System [51] supplies photographs with normative valence and arousal ratings; EMOTIC [39] annotates people in context with 26 categories and continuous dimensions, and shows that scene context around a person changes the emotion read from them. The art resources are closest in spirit to the present corpus because they pair a picture with a short affective text: ArtEmis [1] collected 439,000 utterances explaining the emotion an artwork evokes,

ArtELingo [55] extended it across four languages and found that emotion labels vary with language and culture, and WikiArt Emotions [59] annotated four thousand paintings and reported two findings this project leans on: a title shifts the perceived emotion of an image, and images with faces produce far more consistent emotion judgements than images without.

On the text side, CARER [76] distant-supervised a tweet corpus with eight emotions and is the source of the widely used "emotion" dataset; the SemEval affect tasks of 2018 [58] and 2025 [61] define the shared benchmarks and the intensity-scoring protocol. Human-preference datasets for text-to-image generation, Pick-a-Pic [37], Human Preference Score [82] and ImageReward [83], record which of two generated images people prefer and train scorers on the result; none of them asks about emotion.

For this project the consequence is domain shift. Monochrome line art is out of distribution for every visual-affect resource above, so a classifier trained on any of them cannot simply be applied and trusted. That is one of two reasons the image side describes before it extracts, and it is why the direct-CLIP route is the baseline rather than the method.

2.5 Text-to-image generative models

The drawings were produced by the DALL·E family. The first DALL·E [72] generated images from text with a discrete autoregressive model; its successor [71] conditioned a diffusion decoder on CLIP image embeddings; DALL·E 3 [4] improved prompt-following by training on synthetic captions and, importantly for this corpus, rewrites the user's prompt before generation. The provenance manifest embedded in every image names the software agent only as "DALL·E", without a version, and records no seed and no rewritten prompt. The diffusion lineage that the DALL·E work sits alongside, latent diffusion [73], SDXL [66] and Imagen [75], shares the same documented weaknesses: fine spatial detail, text rendering, and faces.

The vision–language models used as readers here come from the same period. CLIP [68] learned an image–text embedding from 400 million captioned photographs and is the standard zero-shot image classifier, which is exactly why my supervisor advised against asking it for an emotion: it is trained on colour photographs and is strong on objects. BLIP [45] bootstrapped captioning and visual question answering from noisy web data, and its VQA head answers a free-form question about an image; BLIP-2 [44] connects frozen image encoders to large language models through a lightweight bridge. OpenCLIP [34] is the open reimplementation of CLIP used for the baseline.

For this project the generator is a black box whose parameters were never recorded, so nothing can be regenerated and nothing about the model itself can be isolated; the reader models are borrowed instruments, chosen for what they are good at, and the choice of a question-answering describer rather than a classifier follows from Section 2.4.

2.6 Image evaluation: metrics and frameworks

Automatic evaluation of text-to-image output has converged on the question "did the objects, attributes and relations in the prompt appear in the image?" CLIPScore [29] measures the cosine similarity between CLIP embeddings of the image and the text. TIFA [33] generates questions from the prompt and asks a VQA model to answer them from the image; the Davidsonian Scene Graph [16] structures those questions so that dependent ones are only asked when their premises hold; GenEval [25] uses an object detector to check counts, colours and positions; VQAScore [47, 43] asks a single yes/no question of a generative vision–language model and reads off the probability of "yes", and on the GenAI-Bench human ratings it correlates with people far better than CLIPScore. The Conditional Fréchet Distance [38] scores a model's output distribution against a reference conditioned on the prompt. Behind all of these stand the caption metrics that preceded them, SPICE [3] and CIDEr [78], and the distributional FID [30] that scores image quality without reference to the prompt at all.

Two studies say how much to trust any of this. The Holistic Evaluation of Text-to-Image Models [41] found that automatic scores correlate weakly with human judgement on several axes, photorealism and aesthetics among them. Zhang and Wan [90] stress-tested CLIPScore, VQAScore and DSGScore and found that the first and third reverse model rankings when the generation seed changes and shift materially under a one-pixel perturbation, with VQAScore more stable but not immune, on a corpus of about a thousand prompts, roughly the size of the present one.

A second group of studies concerns the judge rather than the score. Chang and colleagues [11] document how skilled prompt artists steer models with style keywords, "line art" and "black and white" among them, and how outputs drift towards a model's average unless steered. Oppenlaender and colleagues [64] report that pose-driven generation shifts the mood of an image unintentionally and that uncanny artefacts alter its emotional reading. Carrasco and colleagues [10] measured how far human raters agree with the emotion labels a generator was steered by, using Krippendorff's alpha and Cohen's kappa, and found agreement far stronger once emotions were collapsed to positive and negative. Horton, White and Iyengar [32] and Millet and colleagues [54] show, across ten experiments, that people rate the same work lower when told it was made by a machine, which is a design constraint on any future human study of these drawings. Two benchmarks probe machine judges directly: Bhattacharyya and Wang [6] found vision–language models largely inept at zero-shot recognition of the emotion an image evokes, with a systematic bias towards positive labels and a sensitivity to the order in which labels are offered; Gao and colleagues [24] found that multimodal models handle valence but not arousal or dominance and conflate the emotion a subject expresses with the emotion a viewer feels.

For this project three decisions follow. None of the alignment metrics measures affect, so none is the instrument, though VQAScore's question-asking design is the closest relative of the describer used here. Their instability is the reason every value in this dissertation is read against a null rather than trusted as a score. And the positive bias and expressed-versus-evoked confusion reported for machine judges are exactly the failure modes the audits in Chapter 4

look for and find.

2.7 The intersection: emotion in generated images

Work that puts emotion into generated images is target-conditioned and coloured. EmoGen [85] generates images from an emotion category through an emotion space aligned to CLIP and evaluates them with a classifier trained on EmoSet; Make Me Happier [46] edits an image so that it evokes a target emotion and reports that positive emotions are harder to separate than negative ones and that fine facial detail is a weakness; EmotiCrafter [20] conditions generation on continuous valence–arousal coordinates; Mehta and Buntain [53] measured whether generators drift towards fear regardless of the prompt, using an image-emotion classifier on one side and a text-emotion classifier on the other, a cross-modal design that is the nearest methodological cousin of the present pipeline. Zhou and colleagues [91] classify the emotion an image evokes by first turning the image into affective text, which mirrors the describe-then-extract architecture from the recognition side.

Alvarado [2] used a diffusion model to render emotion words and extracted colour features from the results, finding that lightness and chroma, more than hue, are the principal colour carriers of emotional meaning in text-to-image output. The study models colour features only; it does not test shape or line, and so says nothing either way about whether form alone can carry affect. That silence is the reason a monochrome corpus is interesting rather than a reason to expect any particular result from it.

The closest prior art is Lomas and colleagues [49], who had 24 raters score 240 images generated by three models from a fixed ten-item emotion typology, each emotion rendered on a person or a robot, on a 0–10 alignment scale. The generating model was the dominant factor, DALL·E 3 far ahead of DALL·E 2 and Stable Diffusion; alignment varied sharply by emotion; and emotions rendered on people scored higher than on robots. Their prompts were templated and their images coloured. Nobody has measured affect agreement on unmediated, mixed-affect self-report rendered without colour, and the repeated prompts in the present corpus allow a consistency measurement that none of these studies could make, since none drew the same words more than once.

2.8 Emotion perception and human judgement

Machine recognition of emotion in images is built on photographs and confounds two constructs. Yang and colleagues [87] trained coupled networks for visual sentiment with weak supervision; EmoSet and the datasets above supply the labels; the benchmarks of Section 2.6 show how fragile the resulting judges are. Underneath sits the distinction that Gao and colleagues [24] formalise and that the supervisor's perception work makes concrete: the emotion a depicted subject expresses is not the emotion a viewer feels, and neither is what a classifier trained on crowd labels returns. The pipeline in this dissertation reads expressed cues, a drawn expression, a posture, an action; a human study would read evoked affect, and the two must

not be run together. The reverse-correlation methods of Jack and Chen [36, 12, 13, 15] are the model for how such a study would be done: measure what observers perceive, do not assume it.

For this project the distinction fixes the language. The image reader's output is called a reading, not a perception, and the single-coder check in Chapter 4 is the first and only point at which a person looks at a drawing.

2.9 Context

The setting was a public festival, and the one paper on how the public receives AI-generated art [70], a CHI study of 565 participants, found that identical artworks are rated lower when merely labelled as machine-made. The drawings in this corpus were shown to their authors as machine-made, and any future human rating of them will have to blind or counterbalance that label.

2.10 The gap

The strands converge on an object none of them has studied. Emotion theory says that short self-reports carry blended, graded affect. Text recognition says that this exact register is where classifiers are weakest. Generation says the models will render such prompts fluently but wobble on faces and detail, and that the prompt they actually render may not be the one the visitor typed. Automatic evaluation says the best metrics score objects, not affect, and are themselves unstable. The affect-in-images work says emotion can be generated and recognised, but chiefly through colour and only on photographs and art, with judges that are biased towards positive labels and confuse expressed with evoked emotion.

Within the 73 papers reviewed here, no work measures the agreement between the emotion recoverable from a short, unmediated, mixed-affect first-person self-report and the emotion recoverable from a colour-free line drawing made from it, and none characterises how stable that reading is when identical words are drawn again. The gap is sharp against each neighbour. Against the compositional metrics, a line drawing can pass "is there a person?" while showing the wrong feeling entirely. Against Lomas and colleagues, the prompts here are real and blended rather than drawn from a typology, and the drawings have no colour. Against the conditional generators, the emotion here is latent in what somebody typed, not a control knob. Against the visual-emotion recognisers, monochrome line art is out of their distribution and none of them separates expressed from evoked affect for such stimuli.

The chapters that follow therefore answer two questions and decline a third. How far do two automatic readers agree on the emotion of a visitor's words and of the drawing made from them, relative to chance, and how does that agreement depend on whether a face can be read? How stable is the reading when the same words are drawn more than once? And, explicitly not answered here because it needs people rather than models: what would a person see?

Chapter 3

Methodology

3.1 Overview and design principles

Three principles run through the design, and each can be traced either to a decision taken in supervision or to something the data forced.

The first is similarity rather than matching. Two readers each return a distribution over the same seven emotion families, and the comparison between them is a ladder of four measures that reward closeness to different degrees, from a continuous distance on the circumplex to an exact top-1 match (Section 3.5). No measure is treated as the score; the ladder is the result.

The second is that every number is read against its own chance level. This was not a principle I started with. It came from noticing, while building the demonstrator, that two records paired at random already score about 76.6% on the loosest measure, because most families sit close together on the circumplex and the measure has a floor of 0.41. A bare value of 0.71 therefore looks like a pass and is below chance. From that point on, every subset and every measure carries a random-pairing null (Section 3.6), every rate carries a clustered confidence interval, and the two contrasts the design is built around are tested formally (Section 3.7).

The third is describe, do not ask. The vision model is never asked what emotion a drawing shows. It answers five factual questions, the answers are composed into one neutral sentence, and a separate text model infers the emotion from that sentence (Section 3.4). The intent is that both sides of the comparison are text read by text models. How much separation this actually buys is not assumed; it is measured by the leak audit and the template ablation in Section 3.8. Figure 3.1 is the map of the chapter.

3.2 The corpus

The corpus is 1,025 records, each a prompt string, a PNG image and an embedded file timestamp. It was collected at the Festival of the Mind in Sheffield between 19 and 29 September 2024. Visitors were invited to describe how they were feeling; the interface

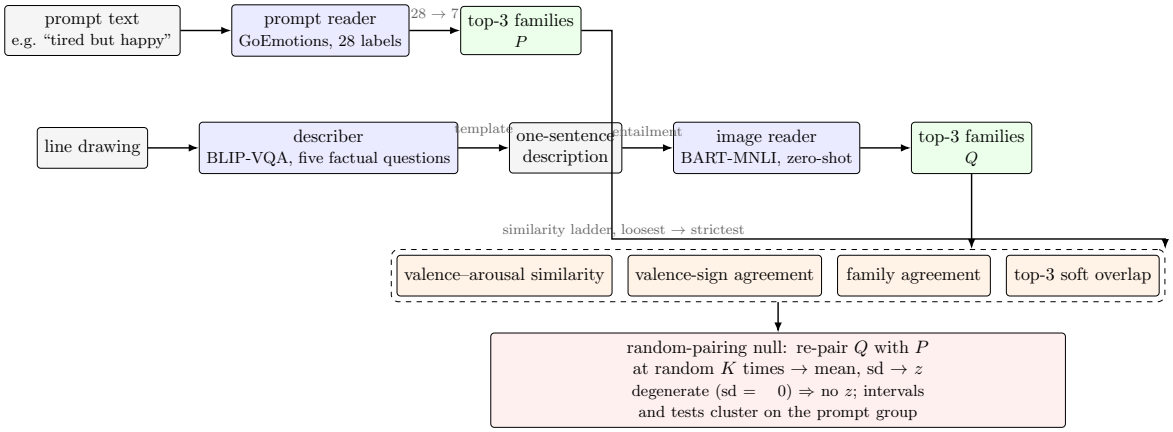


Figure 3.1: The describe-then-extract pipeline and the similarity ladder. The prompt reader (GoEmotions) and the image reader (BART-MNLI over a BLIP-VQA description) each return a top-3 distribution over seven families; four measures compare them, and each measure is read against the mean and spread it takes when the image readings are re-paired with the prompts at random.

accepted any text; each submission produced one drawing. The drawings are 1024 by 1024 pixels and uniformly monochrome, so a line-art style instruction was appended to every prompt by the installation software, and that instruction is not in the data I was given. The provenance manifest embedded in each image names the generator as DALL·E without a version and records neither a seed nor the rewritten prompt that the model is known to construct internally [4], so exact regeneration is impossible and the generation parameters are unknown.

The prompts are short. The median is four words, the mean six, and 293 entries are a single word. About 78% of them are typed entirely in lower case, which matters for name detection later. A few are in Italian, French or German. Records are not linked to individuals: 249 of the 1,012 analysed records repeat a prompt that appears earlier in the corpus, the largest group being the word “happy”, which was drawn 42 times; the number of contributors is unknown and can be no larger than 763. Figure 3.2 shows the composition after pre-processing.

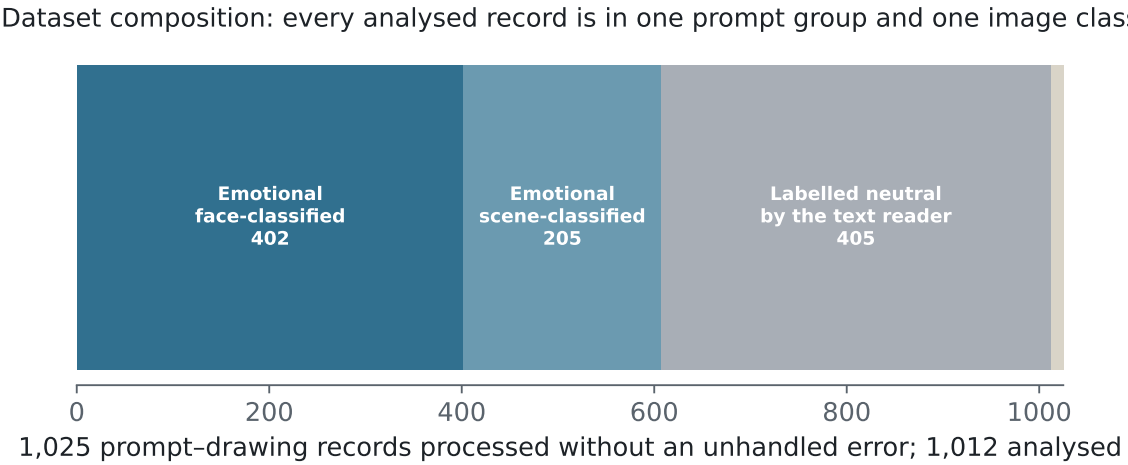


Figure 3.2: Composition of the corpus: 1,025 records processed, 13 pre-festival developer tests excluded, 1,012 analysed. Every analysed record belongs to exactly one prompt group (emotional / neutral-labelled) and one image class (face-classified / scene-classified); the groups are groupings, not exclusions.

[INSERT PHOTOGRAPH OF THE FESTIVAL INSTALLATION]

A photograph of the installation at the Festival of the Mind 2024 (the screen, the pen plotter, or a visitor's drawing being produced), with the source credited in the caption. Optional; delete this box if no photograph is available.

3.3 Data pre-processing

3.3.1 Loading and pre-test removal

The image files follow three incompatible naming patterns (zero-padded, unpadded and prefixed), and two are stored with a lower-case extension that a case-sensitive file system treats as a different file. A naïve loader resolves under a fifth of the corpus. The loader in `emoviz/data.py` handles all three patterns, asserts that exactly 1,025 records resolve, and reads the signing time embedded in each image's provenance manifest, which the data as supplied did not expose. Those timestamps show that records 1 to 13 were created between 26 August and 18 September 2024, before the festival opened, scattered across days in a way consistent with developer testing. They are excluded from analysis but still processed by the

pipeline, so that the completion figure reported later covers all 1,025 images and nothing is silently dropped.

3.3.2 Prompt groups

Every record carries an integer group key computed from its original prompt text, stripped of surrounding whitespace and case-folded, before any redaction is applied. Computing the key before redaction matters: two different people's names both become the same placeholder, and a key computed afterwards would merge two distinct prompts into one group, which happened once during development and was caught by the accounting tests. There are 763 groups; 87 of them hold two or more records, covering 336 records, and 43 of those groups are pairs. Every confidence interval and every permutation test in Section 3.7 resamples or permutes at the level of this key, because records that share a prompt are not independent evidence about the readers. The key is case-insensitive while the prompt reader saw the text as typed, so a handful of groups mix readings ("Ok" and "ok" fall in one group but were read separately); this changes no count and is reported for completeness.

3.3.3 Name redaction

The corpus contains the names of a small number of private individuals, typed by visitors who wrote about a friend, a relative or a teacher. Following my supervisor's direction, these are treated as a data pre-processing matter: names of private individuals are replaced by a placeholder, while content about illness, grief and personal difficulty is kept, because it is not identifying on its own and is essential to the analysis. Redaction happens at load time and the files on disk are never modified, so the analysis, the figures and the demonstrator all see the same redacted text and cannot diverge.

The procedure has three documented stages. Detection is tuned for recall, because with most prompts in lower case capitalisation alone would miss most names: candidates come from the NLTK gazetteer of 7,944 given names [7], from mid-sentence capitalised tokens absent from a 236,000-word English dictionary, and from naming phrases such as "called X" and "my friend X". Review records a decision for every candidate, keep or redact, with a reason, and scopes each redaction to the record it was reviewed in, which is what stops "Jack Frost" and "woolly mammoth" being scrubbed as people; the audit fails if any candidate is unreviewed or if a redaction entry no longer matches anything. Application replaces the token per record and collapses consecutive placeholders to one. Eight records are affected. Re-reading the redacted prompts with the prompt reader moves no family probability by more than 0.012 and changes no top-1 family, so the prompt-side labels are unaffected; two records consisted of a name alone and reduce to the bare placeholder, and the control audit in Section 3.8 checks that they change nothing. The decision table itself names the tokens it removes, so it travels with the dataset rather than with the code.

3.4 The pipeline

3.4.1 Prompt reader

The prompt reader is a RoBERTa model fine-tuned on GoEmotions [21] (`SamLowe/roberta-base-go_emotions`, about 125 million parameters). It returns 28 probabilities, which are summed into the seven families by the grouping the dataset's authors proposed (Appendix B), normalised to a distribution, and the three highest kept. A prompt is called emotional when its top-1 family is not neutral; 607 of the 1,012 analysed prompts are, and 405 are labelled neutral. That label is the reader's judgement and was never human-checked, which is why the neutral-labelled prompts are analysed as a group rather than discarded, and why Section 3.8 audits them.

Two alternatives were tried on the prompt side and rejected with evidence. Zero-shot entailment with the same NLI model used on the image side over-read surprise on short text: a third of prompts came back with surprise as the top family, an artefact of the hypothesis wording on two-word inputs. A hand-built lexicon, the rule used in the first version of the pipeline, was transparent but brittle and could not represent a blended prompt. GoEmotions was retained because its training data are first-person social text, the register closest to a visitor's entry, and because re-extracting the prompt emotions with it left every headline result intact while removing the surprise artefact.

3.4.2 Describer

The describer is BLIP-VQA [45] (`Salesforce/blip-vqa-base`, about 385 million parameters). It answers the five fixed questions in Appendix B, in order: the facial expression, what the person is doing, the body posture, how many people there are, and what is in the picture. The word "emotion" appears in none of them. Answers are cached per record, which is what allows the template ablation to re-run the extraction stage without re-running the vision model.

The describer was the third choice, not the first. A 3-billion-parameter vision–language model was the intended describer and was abandoned after it exhausted the 16 GB of memory on the machine available to me, thrashing into swap for 4 h 45 min with no completed record. A plain caption from BLIP's captioning head was too terse: read for emotion, it collapsed almost every drawing to a single family. BLIP-VQA's expression answer gives the downstream reader a cue that a caption does not, and the run over all 1,025 images completed at between one and twelve seconds a record. That its expression answer does most of the work is the central finding of the audits, and it is stated as such in Chapter 4 rather than hidden in the method.

3.4.3 Description template and image reader

The five answers are composed into one sentence by the template in Appendix B: a drawing of so many people showing such a scene, the person doing this and holding that posture, the facial expression being whatever the describer said. BART fine-tuned on MultiNLI [42, 79]

(`facebook/bart-large-mnli`, about 407 million parameters) then scores each family as a hypothesis, "This scene conveys {family}", by entailment [88], and the three highest scores are kept.

The template has a property that must be said plainly here because everything in Sections 4.8 and 4.10 turns on it. It asserts a person and a facial expression for every drawing, including those in which the describer has just reported that there are no people. It was written that way in the first week of the build and its consequence was not measured until the audits in September. One alternative to the reader was tried and rejected: GoEmotions itself applied to the descriptions collapsed to neutral with probability above 0.95, because it needs first-person feeling words and a description has none.

3.4.4 Face and scene classes

My supervisor's expectation, from the perception literature, was that drawings with a face would be read more reliably than drawings of scenes. A record is face-classified when the describer reports at least one person and its expression answer contains a word from the documented list in Appendix B, matched by token so that "man is smiling" counts; every other record is scene-classified. The people count is parsed from digits and number words. Over all analysed records 591 are face-classified and 421 scene-classified; within the emotional prompts the split is 402 to 205. The class is defined from the same description the image reader reads, and so the face/scene contrast is partly circular by construction. The circularity is stated now, quantified in Chapter 4, and addressed by the single-coder check, which classifies a sample of drawings without reference to the describer at all.

3.5 The similarity ladder

Let each reader return a top-3 list $T = \{(\ell_i, s_i)\}_{i=1}^{3}$ of families $\ell_i \in \mathcal{F}$ with scores s_i, where $\mathcal{F} = \{\text{joy}, \text{sadness}, \text{anger}, \text{fear}, \text{surprise}, \text{disgust}, \text{neutral}\}$. Write P for the prompt reader's list and Q for the image reader's, and $\text{top}(T) = \ell_1$ for the top-1 family. Each family has a fixed centroid $c_\ell = (v_\ell, a_\ell)$ on the valence–arousal circumplex (Appendix B), hand-placed from the standard positions of the basic emotions [74]; per-family means from the NRC valence–arousal norms [56] are the documented alternative and a further-work item. The four measures below are ordered from loosest to strictest, and each rewards something the others ignore.

Valence–arousal similarity. The score-weighted centroid of a reading and the similarity between two readings are

$$\mathbf{v}(T) = \frac{\sum_i s_i \, c_{\ell_i}}{\sum_i s_i}, \qquad S_{\text{VA}}(P, Q) = 1 - \frac{1}{\sqrt{2}} \left\| \left(\tfrac{v_P+1}{2} - \tfrac{v_Q+1}{2},\; a_P - a_Q \right) \right\|_2, \tag{3.1}$$

where $(v_P, a_P) = \mathbf{v}(P)$ and likewise for Q. Valence is rescaled to $[0, 1]$ so both axes have unit range. This is the measure that gives partial credit to a near miss: a prompt read as joy and a drawing read as surprise are close on the plane, and the measure says so. It is also

the measure that cannot fall to zero. Over the seven centroids the smallest attainable value is 0.412 (joy against sadness), and randomly paired records average about 76.6%; this floor is why the measure is never quoted without its null.

Top-3 soft overlap. Each side is first normalised to a distribution over its own three families, $\tilde{p}_\ell = s_\ell / \sum_j s_j$, so the prompt reader's multi-label scale and the image reader's softmax scale are comparable; the overlap is the sum of per-family minima,

$$S_{\mathrm{ovl}}(P, Q) = \sum_{\ell \in \mathcal{F}} \min\big(\tilde{p}_\ell,\ \tilde{q}_\ell\big) \in [0, 1], \tag{3.2}$$

which equals one only when the two distributions coincide. It rewards a second or third choice that the other reader also made, and it is the only measure that uses the scores rather than just the labels.

Family agreement. Over a set of n records,

$$A_{\mathrm{fam}} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\big[\mathrm{top}(P_i) = \mathrm{top}(Q_i)\big]. \tag{3.3}$$

This is the strictest rung and the one most comparable with other studies; it is also the one whose chance level depends most on the family marginals, since a reader that says joy often will agree with a prompt reader that says joy often.

Valence-sign agreement. With $\sigma(\ell) \in \{+, -, 0\}$ the sign of a family (+ for joy; − for sadness, anger, fear, disgust; 0 for surprise and neutral),

$$A_{\mathrm{sign}} = \frac{\sum_i \mathbb{1}\big[\sigma(\mathrm{top}\, P_i) \neq 0 \ \wedge\ \sigma(\mathrm{top}\, P_i) = \sigma(\mathrm{top}\, Q_i)\big]}{\sum_i \mathbb{1}\big[\sigma(\mathrm{top}\, P_i) \neq 0\big]}, \tag{3.4}$$

so records whose prompt family has no sign are excluded from the denominator; the denominator is reported with every value (for example 530 of the 607 emotional prompts). It asks the coarsest categorical question, whether the two readers landed on the same side of neutral, and it is the rung on which the headline result rests.

3.6 The random-pairing null

The measures do not start at zero. Equation 3.1 has a floor, and the family marginals in Table 4.11 make agreement on joy likely by chance alone. The null adopted is the exchangeability hypothesis that the image reading is unrelated to the prompt [26]: if that were true, re-pairing the drawings with the wrong prompts would change nothing on average. For a subset of n records and a measure m, draw K random permutations π_k of the image readings and compute $m_k = m\big(\{(P_i, Q_{\pi_k(i)})\}_{i=1}^{n}\big)$. With $\hat{\mu} = \frac{1}{K}\sum_k m_k$ and $\hat{\sigma}$ the standard

deviation of $\{m_k\}$,

$$z = \frac{m_{\text{obs}} - \hat{\mu}}{\hat{\sigma}}, \qquad K = 300, \text{ seed } 7. \tag{3.5}$$

Two cautions attach to z. With $K = 300$ re-pairings, $\hat{\sigma}$ is itself an estimate; across seeds 1 to 5 the headline z moves between +11.2 and +12.2 while the null mean stays within a tenth of a point, so z orders effects and should not be read as a precise probability. And a null can be *degenerate*: when $\hat{\sigma} = 0$, which happens for family agreement when every prompt in the subset shares one top-1 family and for valence-sign agreement when every prompt shares one sign, permuting the image side cannot change the score and there is nothing to test. The analysis detects this from the null itself and prints no z for such cells. An earlier version flagged degeneracy by hand for two subsets and missed a third; detecting it from the null's spread removed the judgement.

3.7 Inference

Records are not independent. 249 of them repeat an earlier prompt, and the drawings of one prompt share whatever that prompt does to the readers. Every interval and every test below therefore treats the prompt group as the unit [9].

Clustered bootstrap confidence intervals. Let $\mathcal{G}$ be the set of prompt groups in a subset. For $b = 1, \ldots, B$ draw $|\mathcal{G}|$ groups with replacement, pool their records, and recompute the measure; the 95% interval is the 2.5th and 97.5th percentiles of the B values [22], with $B = 10,000$ and seed 7.

Clustered permutation test of the face/scene difference. Under the null that the image class is unrelated to agreement, the face/scene labels are exchangeable across prompt groups. The labels are permuted across groups P times and the difference in agreement recomputed; the two-sided p-value uses the estimator of [63],

$$p = \frac{k+1}{P+1}, \qquad k = \#\{\text{permuted differences at least as extreme as the observed}\}, \tag{3.6}$$

so the smallest reportable value with $P = 10,000$ is 10^{-4}.

Cohen's kappa. For comparability with other studies, family agreement is also reported as

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{3.7}$$

with p_o the observed top-1 agreement and p_e the agreement expected from the two readers' family marginals [18], interpreted with the bands of [40]; intervals by the clustered bootstrap above.

Multiple comparisons. The analysis prints 36 non-degenerate z-scores. Each is converted to a two-sided p by the normal approximation and the family is controlled with the Holm–Bonferroni step-down procedure [31]: order $p_{(1)} \leq \cdots \leq p_{(m)}$ and reject while $p_{(i)} \leq \alpha/(m - i + 1)$, $\alpha = 0.05$. The normal approximation is nominal for the largest z; it affects none of the decisions, which turn on the cells near the threshold.

Paired comparison of two image readers. On the same records each reader either agrees with the prompt reader or does not. With b the records only the first reader agrees on and c only the second, McNemar's exact binomial test [52] on (b, c) treats records as independent and is therefore optimistic here; the clustered version flips the sign of each prompt group's summed difference as a block (20,000 draws) and the clustered bootstrap gives the interval on the difference in rates.

Wilson intervals. Proportions from the single-coder check use the Wilson score interval [80], which behaves at small n and near 0 and 1.

3.8 Analyses

The analyses are listed in the order Chapter 4 reports them; each is one script, and each writes a JSON file from which the tables, the figures and the numbers in this document are generated.

Splits and the main table. `analysis_final.py` computes the four measures and their nulls on the emotional prompts, on the face- and scene-classified records within them, on the positive- and negative-valence prompts (where only the valence–arousal measure has a live null), and on the neutral-labelled prompts, all against Equation 3.5. The same table is recomputed on the first occurrence of each prompt, 763 records, as the robustness check against repetition.

The face/scene difference. `significance.py` computes the clustered bootstrap interval on the difference and the clustered permutation test above, for valence-sign and family agreement. `inference.py` adds the clustered intervals for every subset and measure, Cohen's kappa, the Holm procedure over every printed z, and the seed sensitivity of the headline z.

Specificity check. Prompts the text reader labels neutral should show no agreement above their null. Only the valence–arousal measure is informative there, since the other three are degenerate on a subset whose every prompt shares one label.

Method comparison. `method_comparison.py` holds the prompt reader, the subset, the four measures and the random re-pairings fixed and varies only the image reader: describe-then-extract against a direct CLIP zero-shot reader [68, 34] (ViT-B/32, LAION-2B weights) prompted

with the seven families, its output frozen in a results file so the main pipeline cannot overwrite it. The paired tests are those of Section 3.7. An earlier comparison from the first build used a different prompt reader, a different subset and a different overlap formula; it is kept as a record in Table 4.7 and is not comparable with the main analysis.

Consistency across repeated prompts. For a prompt group g with $n_g \geq 2$ records and image top-1 families $\{\text{top}(Q_i)\}_{i \in g}$, the modal fraction is $c_g = \max_\ell n_{g,\ell}/n_g$, and the consistency is the mean $C = \frac{1}{|\mathcal{G}_{\geq 2}|} \sum_g c_g$ over the 87 groups. Because 43 groups are pairs, c_g cannot fall below 0.5 for them, so C is read against a permutation null that shuffles the image top-1 families across all 1,012 analysed records and re-cuts them into the same groups (10,000 draws). The count of fully consistent groups ($c_g = 1$) is treated the same way. This bounds the reproducibility of the whole prompt-to-reading chain; it does not isolate the generator, whose parameters were not recorded, and it cannot separate a visitor who typed "happy" twice from two visitors who typed it once each.

Control audit. `audits.py` runs a fixed lexicon of 66 plain English affect words over the 405 neutral-labelled prompts, identifies the two records that reduce to the placeholder, and re-runs the specificity check on the cleaned control with both sets removed.

Describer audit and copy rate. The same script crosses the describer's people count with the presence of an expression word, counts the smile answers, and computes the copy rate. Where the describer's expression answer is a single word that names an emotion (strict: happy, sad, angry, ...; 418 records) or a facial action with an implied family (broad: smile, crying, ...; 997 records), the copy rate is the share of those records in which the image reader's top-1 family equals the family the word implies. It measures how much the "separate" extraction stage adds beyond the describer's own word.

Error taxonomy. `error_taxonomy.py` selects every emotional record on which both readers scored their top-1 family at 0.8 or above and disagreed, tags each automatically (people count, bare expression word, mixed-feeling connective, ambiguous raw label, short prompt), and then, for each, I looked at the drawing and assigned one primary failure mode from a fixed set of six: the describer invented a face or expression; the describer misread one; the reader mis-inferred from a correct description; the prompt was ambiguous or mixed; the drawing does not depict the prompt; the prompt reader was wrong. The tie-break rule is written in the script's output.

Template ablation. `ablate_nli.py` rebuilds each record's description from its cached answers under an altered template, re-runs only the entailment stage, and recomputes the ladder and its nulls with the same code and seed. The variants drop one clause at a time (expression, action, posture, people count, scene), fix the expression word to "unclear" for every record, and, in two people-aware variants, omit the person and expression clauses

wherever the describer reported no people. A control variant re-runs the unmodified template under the same conditions, so numerical noise can be separated from template effects.

Single-coder check. `human_check.py` draws a fixed sample of 120 drawings stratified on the describer's output (face-classified, scene-classified with no people reported, and every record with a vague or word-only people answer), lays them out as numbered contact sheets that show nothing but the drawings, and, once coded, scores the describer's people count and expression answer and the image reader's valence against the coder with Wilson intervals, then re-runs the face/scene contrast on the coder's split. The coder works blind to the prompt, the description and both readings. Figure 3.3 shows one sheet.



Figure 3.3: The first of six contact sheets for the single-coder check. Each drawing is identified only by its sheet number; the prompt, the description and the readers' outputs are withheld during coding.

3.9 The demonstrator

The pipeline is exposed as a small local web application (`app.py`) that runs the real models on any of the 1,012 records or on an uploaded image, shows the five answers and the composed description, the two readings side by side, and the four measures against the same null the analysis uses, computed at start-up from the same result files. Three design choices carry the methodology into the interface. The valence–arousal axis starts at its floor rather than at zero and shades the middle half of random pairings, so a value that is below chance looks below chance. The verdict leads with valence-sign and family agreement and reports the loosest measure as a percentile of random pairings. And where a chance level is degenerate the interface says so rather than printing one. Nothing a visitor types or uploads is stored, and the request log is suppressed. It was built as a testing and communication tool, used in the supervision meetings and the project presentation, and it is not a data-collection instrument.

[INSERT SCREENSHOT OF THE DEMONSTRATOR]

A screenshot of the demonstrator after running record 63 ("super tired"): the drawing, the five answers, the two readings and the verdict against the chance band. Capture it with `bash emoviz/run_interface.sh`, choose record 63, and press Compare.

3.10 Reproducibility and testing

The environment is pinned (`requirements.txt`; Python 3.12, PyTorch 2.13, Transformers 5.14), every random process is seeded, and every model output is cached per record, so that every analysis after the pipeline itself re-runs in seconds. The direct-CLIP baseline is a frozen file. One module, `emoviz/metrics.py`, holds the only definition of every constant, measure, split rule and the null, and every script imports it; before it existed two scripts carried two slightly different overlap formulas, and the difference, though negligible on this data, was exactly the kind of drift the module now prevents.

Testing has two layers. `test.sh` runs 38 smoke checks: the environment, the integrity of the corpus and the cached outputs, the redaction audit, every analysis script to a clean exit, a live three-image run of the pipeline on fresh images, and the figure build. `test_numeric.py` pins 54 values: every headline number in this document is asserted against the JSON file that produces it within a stated tolerance, and the measures themselves are unit-tested (the 0.412 floor, symmetry, the bounds and scale invariance of the overlap, the sign map, the face rule). A change in the analysis fails the build rather than drifting into the text. The figures (`make_figures.py`) and every table and number in this dissertation (`export_tex.py`) are

generated from the same files, which is why no result number is typed by hand anywhere in the document.

What the tests do not do should be stated as plainly. None of them validates a model answer against a human judgement. The pipeline's completion figure, all 1,025 images processed with no unhandled error, is a robustness statement and not a correctness one; the single-coder check is the only place a person enters the chain. Appendix A lists the files and the run order.

Chapter 4

Results

This chapter reports what the analyses found. Interpretation is held back for Chapter 5; here the tables and figures carry the numbers and the prose says what each shows. Every value is a macro generated from the result files, every rate is read against its random-pairing null (300 re-pairings, seed 7), and every interval is a clustered bootstrap over prompt groups (10,000 draws).

4.1 Overall agreement

Table 4.1 gives the four measures on the 607 prompts the text reader labels emotional, and Figure 4.1 shows the same cells as distances from their nulls. Every measure sits well above chance. Valence-sign agreement is 64.0% against a null of 45.6% ($z = +11.1$, 95% CI [58.6, 68.8]); the denominator is 530 rather than 607, because prompts whose top family is surprise have no valence sign. Family agreement is 53.0% against 38.8% ($z = +10.3$, CI [47.3, 58.7]), top-3 overlap 50.6% against 39.4% ($z = +11.6$), and valence–arousal similarity 83.8% against 77.3% ($z = +12.2$). The loosest measure has the largest z and the smallest gap in percentage points, which is the floor of Equation 3.1 at work: its null is high and tight, so a modest excess over it is many standard deviations.

4.2 Face-classified versus scene-classified drawings

Figure 4.2 and Table 4.2 split the emotional prompts by the describer's image class. On the 402 face-classified records valence-sign agreement is 73.9% against a null of 53.5% ($z = +10.4$, CI [68.3, 79.2]) and family agreement 61.9% against 45.8% ($z = +10.3$). On the 205 scene-classified records the same measures are 42.9% against 29.6% ($z = +5.8$, CI [33.1, 51.1]) and 35.6% against 26.2% ($z = +4.4$). Both classes clear their nulls in this table; Section 4.10 returns to the scene row.

The two nulls differ, so the contrast is stated as a difference rather than a ratio. Face minus scene is +30.9 pp on valence-sign agreement (CI [+22.6, +40.6]) and +26.3 pp on family agreement (CI [+18.4, +35.3]), with the clustered permutation test giving $p = 0.0001$

Table 4.1: Agreement between the prompt reader and the image reader on the 1,012 analysed records, by prompt group and image class, against the random-pairing null (300 re-pairings, seed 7). CI: clustered bootstrap over prompt groups (10,000 draws). n/a: degenerate null (every prompt in the subset shares one label, so permutation cannot change the score).

Subset / measure	Observed	Null	z	95% CI
Emotional prompts (n=607; sign n=530)				
Valence–arousal similarity	83.8%	77.3%	+12.2	[82.0, 85.5]
Valence-sign agreement	64.0%	45.6%	+11.1	[58.6, 68.8]
Family agreement	53.0%	38.8%	+10.3	[47.3, 58.7]
Top-3 soft overlap	50.6%	39.4%	+11.6	[46.5, 54.7]
face-classified (n=402; sign n=360)				
Valence–arousal similarity	84.2%	76.4%	+11.2	[82.3, 86.1]
Valence-sign agreement	73.9%	53.5%	+10.4	[68.3, 79.2]
Family agreement	61.9%	45.8%	+10.3	[55.8, 68.0]
Top-3 soft overlap	53.4%	40.4%	+11.6	[49.2, 57.7]
scene-classified (n=205; sign n=170)				
Valence–arousal similarity	82.9%	79.1%	+6.0	[80.4, 85.2]
Valence-sign agreement	42.9%	29.6%	+5.8	[33.1, 51.1]
Family agreement	35.6%	26.2%	+4.4	[26.4, 44.1]
Top-3 soft overlap	44.9%	38.1%	+4.9	[38.7, 51.0]
Neutral-labelled prompts (n=405; sign n=0)				
Valence–arousal similarity	76.6%	76.6%	+0.3	[75.6, 77.6]
Valence-sign agreement	0.0%	0.0%	n/a	degenerate
Family agreement	28.6%	28.6%	n/a	degenerate
Top-3 soft overlap	31.6%	31.6%	+0.0	–

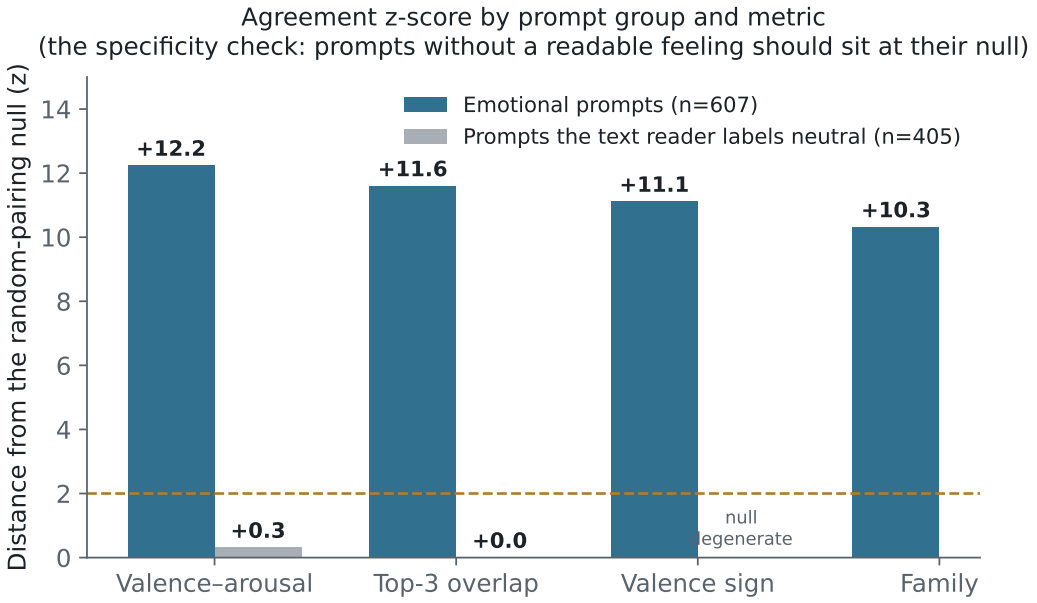


Figure 4.1: Distance from the random-pairing null (z) by prompt group and measure. For the neutral-labelled prompts the sign and family nulls are degenerate and no z is shown.

on both. Valence–arousal similarity does not separate the classes (84.2% against 82.9%); the difference lives in the categorical rungs.

Table 4.2: Face-classified versus scene-classified drawings on the emotional prompts: agreement with 95% clustered-bootstrap CIs, the difference with its CI, and the clustered permutation p (10,000 draws; significance.py).

Measure	Face (n=402)	Scene (n=205)	Difference (pp)	Perm. p
Valence-sign agreement	73.9% [68.3, 79.2]	42.9% [33.1, 51.1]	+30.9 pp [+22.6, +40.6]	0.0001
Family agreement	61.9% [55.8, 68.0]	35.6% [26.4, 44.1]	+26.3 pp [+18.4, +35.3]	0.0001

4.3 Robustness to repeated prompts and multiple comparisons

Table 4.3 recomputes the main table on the first occurrence of each prompt, 763 records of which 452 are emotional. The overall and face results hold: on the 319 distinct face-classified records valence-sign agreement is 71.8% ($z = +8.4$) and family agreement 59.9% ($z = +8.1$). The scene result weakens substantially: valence-sign agreement falls to 37.3% ($z = +3.5$) and family agreement to 30.8% ($z = +2.6$) on 133 records.

Table 4.4 applies the Holm–Bonferroni procedure to the 36 non-degenerate z-scores the analysis prints. 26 survive at family-wise 0.05 and 10 do not. Every full-sample cell on the emotional, face and scene subsets survives, as does every distinct-prompt cell on the emotional and face subsets. The cells that fail are the ones expected to sit at chance (the neutral-labelled

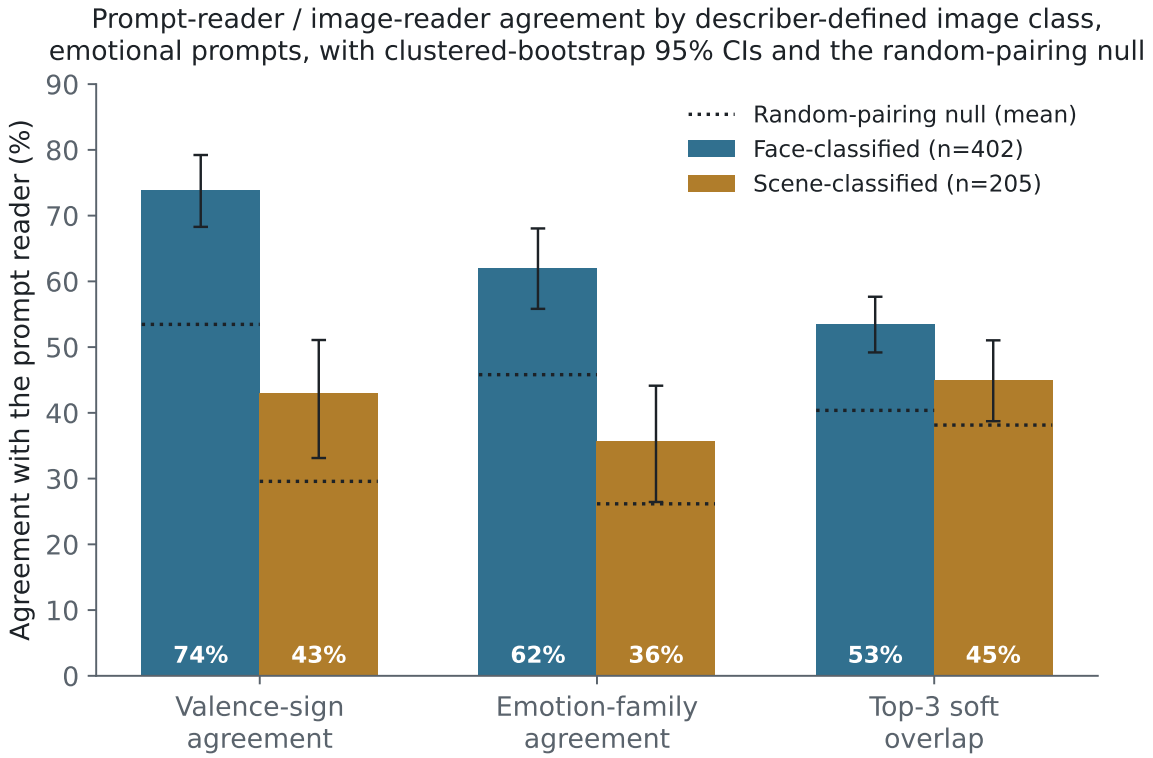


Figure 4.2: Agreement between the prompt reader and the image reader on the emotional prompts, by describer-defined image class, with clustered-bootstrap 95% CIs (error bars) and each cell's random-pairing null (dotted).

control on both blocks), the single-valence subsets' valence–arousal and overlap cells, and the distinct-prompt scene family cell ($z = +2.6$), which should be read as suggestive rather than established.

Table 4.3: The same analysis on the 763 distinct prompts only (first occurrence of each prompt text kept), a check against pseudo-replication from the 249 repeated prompts.

Subset / measure	Observed	Null	z	95% CI
Emotional prompts (n=452; sign n=394)				
Valence–arousal similarity	83.0%	76.5%	+10.6	[81.5, 84.4]
Valence-sign agreement	62.2%	44.4%	+9.9	[57.2, 67.0]
Family agreement	51.3%	37.2%	+9.3	[46.7, 56.0]
Top-3 soft overlap	48.9%	37.7%	+10.8	[45.9, 51.9]
face-classified (n=319; sign n=284)				
Valence–arousal similarity	83.5%	75.5%	+9.7	[81.7, 85.2]
Valence-sign agreement	71.8%	51.0%	+8.4	[66.4, 77.0]
Family agreement	59.9%	43.0%	+8.1	[54.2, 65.2]
Top-3 soft overlap	51.8%	38.4%	+9.7	[48.1, 55.4]
scene-classified (n=133; sign n=110)				
Valence–arousal similarity	81.8%	78.9%	+3.6	[79.5, 83.8]
Valence-sign agreement	37.3%	27.1%	+3.5	[28.1, 46.2]
Family agreement	30.8%	24.1%	+2.6	[23.3, 38.4]
Top-3 soft overlap	41.7%	36.7%	+3.1	[36.9, 46.5]
Neutral-labelled prompts (n=311; sign n=0)				
Valence–arousal similarity	76.6%	76.4%	+1.1	–
Valence-sign agreement	0.0%	0.0%	n/a	degenerate
Family agreement	27.7%	27.7%	n/a	degenerate
Top-3 soft overlap	31.4%	31.1%	+0.8	–

4.4 Specificity check: prompts the text reader labels neutral

On the 405 prompts the text reader labels neutral, only the valence–arousal measure has a live null. It sits on it: 76.6% observed against 76.5% expected ($z = +0.5$ at record level, -0.1 against a null clustered on prompt groups). Family agreement equals its null exactly (28.6%) because every prompt in the subset shares the neutral label, so permutation cannot move it; the same holds for valence sign, whose numerator and denominator are both zero here.

The control audit (Table 4.5) qualifies the set. 68 of the 405 prompts (16.8%) contain a plain English affect word from a fixed 66-word lexicon, most often calm (10), content (8), tired (7), stress (6), relaxed (6), inspired (4), sleepy (4), interested (3), anticipation (3), nostalgic (3); 2 consist of a redacted name alone. On the cleaned control of 335 records the result is unchanged: 76.6% against 76.6% ($z = +0.0$). The null's spread on this subset is very small (0.15%) because the prompt-side points barely vary, so observed and expected values are expected to coincide here; the check is passed, and it is a weak one.

Table 4.4: Multiple-comparison control over the 36 non-degenerate z-scores the analysis prints (12 degenerate ones excluded). Two-sided p by the normal approximation; Holm–Bonferroni at family-wise 0.05: 26 survive, 10 do not.

Outcome	Tests
Do not survive	DISTINCT negative prompts overlap z+2.7; DISTINCT scene images family z+2.6; FULL positive prompts overlap z+2.3; FULL negative prompts va z+2.1; DISTINCT positive prompts overlap z+1.9; DISTINCT negative prompts va z+1.6; DISTINCT non-emotional va z+1.1; DISTINCT non-emotional overlap z+0.8; FULL non-emotional va z+0.3; FULL non-emotional overlap z+0.0
Survive	the remaining 26, including every headline (emotional, face and scene on all four measures; distinct-prompt emotional and face)

Table 4.5: The specificity check on the prompts the text reader labels neutral. Only valence–arousal similarity is informative there (sign and family are degenerate: every prompt shares the neutral label). Cleaned control = minus the 2 records reduced to '[name]' and the 68 containing one of 66 plain affect words.

Control set	n	VA observed	VA null	z (record null)	z (clustered null)
All neutral-labelled prompts	405	76.6%	76.5%	+0.5	-0.1
Cleaned control	335	76.6%	76.6%	+0.0	-0.6

4.5 Two image readers, like for like

Figure 4.3 and Table 4.6 score two image readers against the same prompt reader on the same 607 emotional records with the same measures and the same random re-pairings. Describe-then-extract agrees with the prompt reader on 53.0% of families against 22.4% for direct CLIP, a difference of +30.6 pp (CI [+25.5, +35.6]); the cluster sign-flip test gives p <0.001, and the unclustered exact McNemar test, which treats records as independent and is optimistic, gives $p = 3.4 \times 10^{-35}$. On valence sign the figures are 64.0% against 31.1% (+32.8 pp, CI [+27.4, +38.2]); on overlap 50.6% against 26.4%; on valence–arousal similarity 83.8% against 73.3%. Of the 607 records, 218 are ones only describe-then-extract agrees on and 32 ones only direct CLIP agrees on; 104 are agreed by both and 253 by neither.

Two details in the table need saying. Direct CLIP's family z (+11.1) is higher than describe-then-extract's (+10.3) although its rate is far lower, because its null is lower and tighter: it reads 283 of the 607 images as neutral, so it rarely agrees by chance either. The z measures distance from chance, not level of agreement, and the paired tests on the rates are the comparison. And the two image readers agree with each other on only 32.1% of top-1 families. Table 4.7 reproduces the comparison from the first build (608 records, an entailment reader on the prompt side, a different overlap formula), kept as a record; it is not comparable with the main analysis and its figures are not quoted.

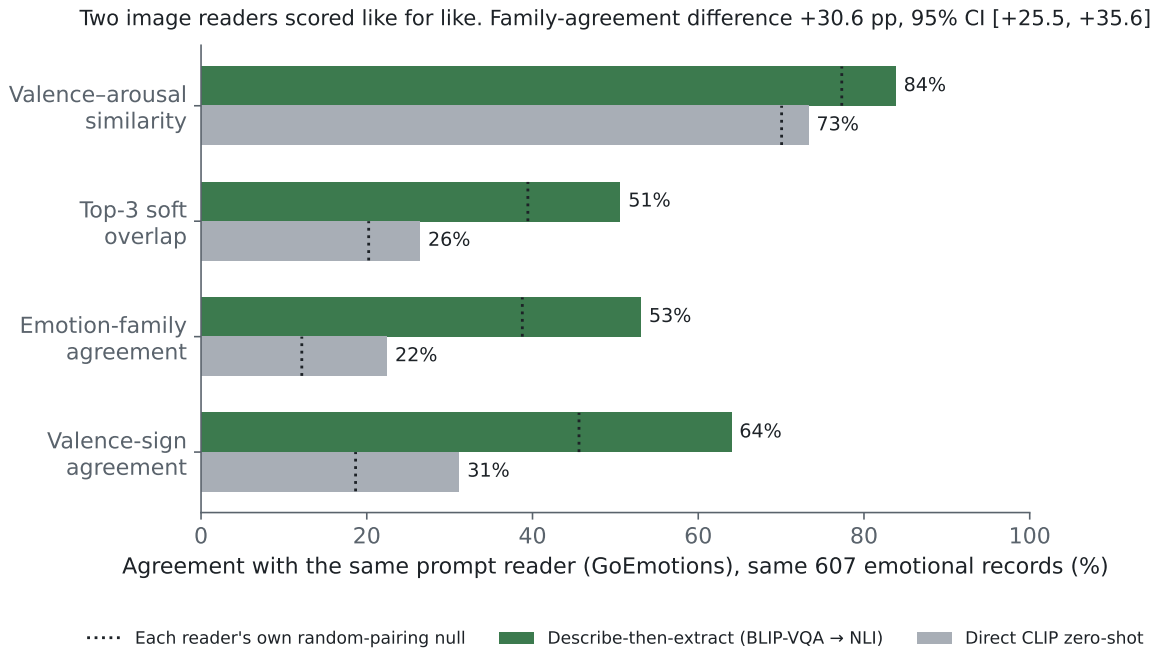


Figure 4.3: Two image readers scored against the same prompt reader on the same 607 emotional records, each against its own random-pairing null (dotted).

Table 4.6: Two image readers scored like for like against the same prompt reader (GoEmotions) on the same 607 emotional records with the same measures and the same 300 re-pairings. b = records only describe-then-extract agrees on; c = records only direct CLIP agrees on. The exact McNemar p treats records as independent and is optimistic (3.4×10^{-35} for family); the cluster sign-flip randomisation (20,000 draws) respects prompt groups.

Measure	Describe→extract	null	z	Direct CLIP	null	z
Valence–arousal similarity	83.8%	77.3%	+12.2	73.3%	70.1%	+10.2
Valence-sign agreement	64.0%	45.6%	+11.1	31.1%	18.6%	+9.9
Family agreement	53.0%	38.8%	+10.3	22.4%	12.2%	+11.1
Top-3 soft overlap	50.6%	39.4%	+11.6	26.4%	20.2%	+11.2
Paired family test	b=218, c=32			diff +30.6 pp	CI [+25.5, +35.6]	cluster $p < 0.001$
Paired valence-sign test	b=204, c=30			diff +32.8 pp	CI [+27.4, +38.2]	cluster $p < 0.001$

Table 4.7: The earlier (v1) comparison as published in the build log: prompt side read by BART-MNLI, the 608 records of the v1 lexicon rule, and a different overlap formula. Kept as a record; not comparable with Table 4.1.

Measure	Describe→extract	Direct CLIP
Family agreement	49.3%	25.3%
Top-3 soft overlap	48.6%	22.8%
Valence–arousal similarity	83.7%	74.2%

4.6 Consistency across repeated prompts

87 prompts were drawn two or more times, covering 336 records; 43 of the groups are pairs. Figure 4.4 and Table 4.8 give the within-group agreement of the image reader. The mean share of drawings in a group that receive the group's modal family is 68.8% (CI over groups [63.8, 73.9]) against a permutation null of 59.4% (sd 2.1%), $z = +4.5$, permutation $p = 1.0 \times 10^{-4}$. The same valence sign is returned for 74.5% of drawings within a group against 63.7% ($z = +5.0$). 28 of the 87 groups (32.2%) are fully consistent, every drawing receiving one family, against 14.5 expected ($z = +4.2$). The largest group, the word "happy" drawn 42 times, was read as joy 34 times, neutral 7 times and surprise 1 once.

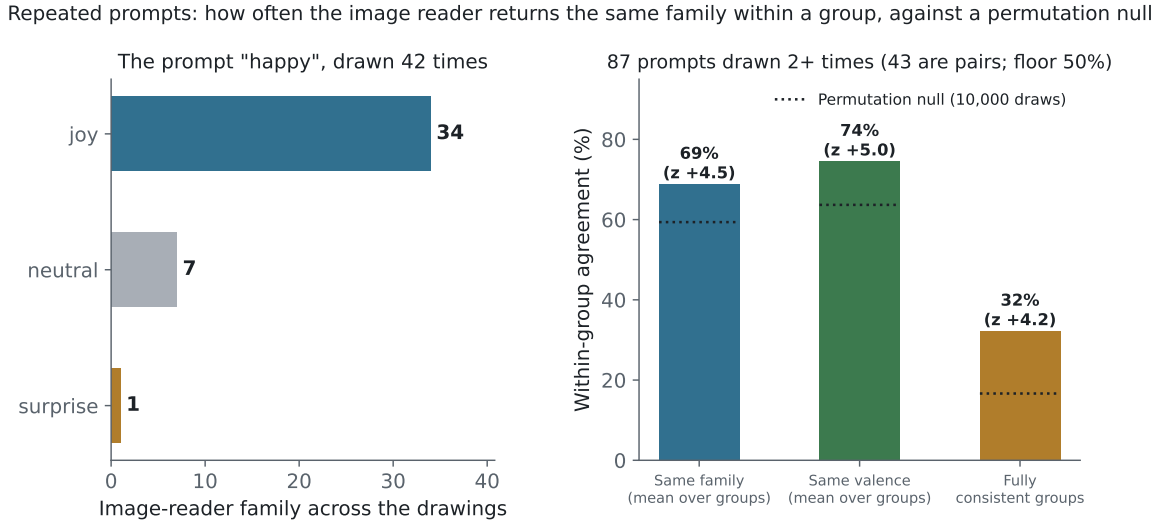


Figure 4.4: Repeated prompts: the image reader's top-1 family across the 42 drawings of "happy" (left), and within-group agreement over all 87 repeated prompts against the permutation null (right).

Table 4.8: Within-group agreement of the image reader across the 87 prompts drawn two or more times (336 records; 43 groups are pairs, whose modal agreement cannot fall below 50%), against a permutation null that shuffles the image readings over all 1,012 analysed records and re-cuts them into the same groups (10,000 draws, seed 7).

Quantity	Observed	Null mean (sd)	z	p
Same family, mean over groups	68.8%	59.4% (2.1%)	+4.5	1.0×10^{-4}
Same valence, mean over groups	74.5%	63.7% (2.1%)	+5.0	1.0×10^{-4}
Fully consistent groups (of 87)	28	14.5 (3.2)	+4.2	2.0×10^{-4}

4.7 Per-family agreement

Figure 4.5 and Table 4.9 cross the prompt reader's family with the image reader's on the 607 emotional records; Table 4.11 gives the marginals over all 1,012 analysed records. The prompt side is dominated by joy (402 of the emotional prompts), with sadness (99) and surprise (77) next and fear (16), anger (12) and disgust (1) rare. Recall by prompt family, meaning the share of prompts in a family whose drawing the image reader placed in the same family, is 64.9% for joy (CI [58.7, 70.3]), 40.4% for sadness (CI [30.9, 51.2]), 50.0% for anger on 12 prompts, 19.5% for surprise (CI [11.4, 28.1]), and 0.0% for fear on 16 prompts. Across all 1,012 records the image reader never returns disgust and returns fear 4 times.

Cohen's kappa for family agreement (Table 4.10) is 0.23 on the emotional prompts (CI [0.18, 0.29]), 0.30 on the face-classified records and 0.13 on the scene-classified: fair, fair and slight on the Landis–Koch scale. In practice the seven-family ladder is a three-family one, joy, sadness and neutral, with anger a distant fourth; the remaining families are too rare on the prompt side, or never produced on the image side, to support any claim.

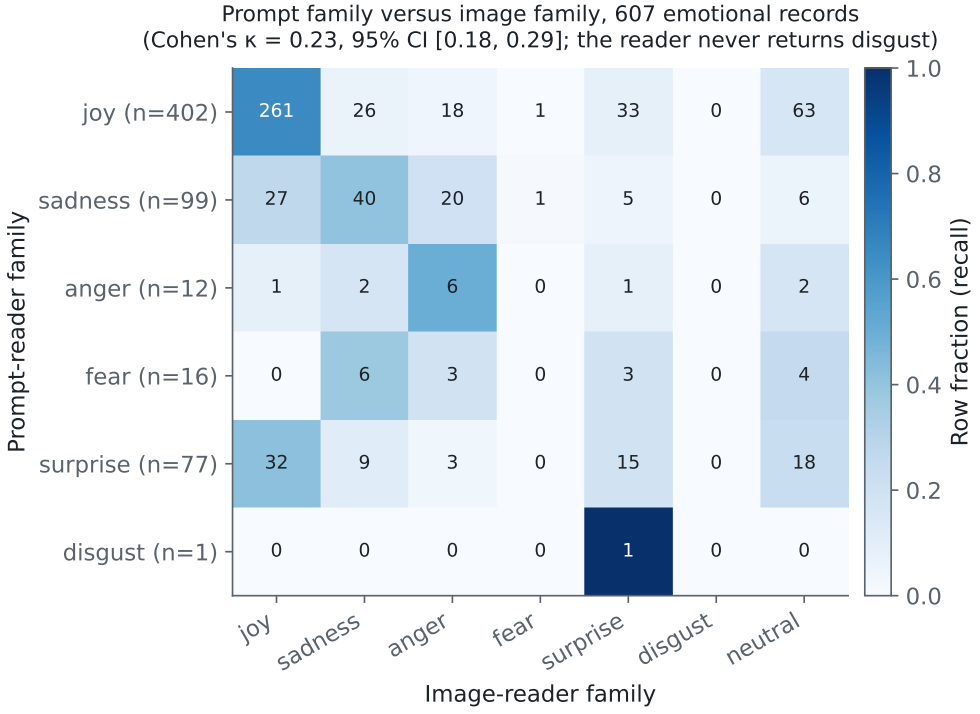


Figure 4.5: Prompt-reader family (rows) against image-reader family (columns) on the 607 emotional records; cell = count, shading = row fraction.

Table 4.9: Prompt-reader family (rows) against image-reader family (columns) on the 607 emotional records. Recall = row diagonal share, with a clustered-bootstrap 95% CI. Cohen's $\kappa = 0.23$ [0.18, 0.29]. The image reader never returns disgust and returns fear 4 times in 1,012 records.

Prompt \ image	joy	sadness	anger	fear	surprise	disgust	neutral	n	Recall [CI]
joy	261	26	18	1	33	0	63	402	64.9% [59, 70]
sadness	27	40	20	1	5	0	6	99	40.4% [31, 51]
anger	1	2	6	0	1	0	2	12	50.0% [22, 78]
fear	0	6	3	0	3	0	4	16	0.0% [0, 0]
surprise	32	9	3	0	15	0	18	77	19.5% [11, 28]
disgust	0	0	0	0	1	0	0	1	0.0% [0, 0]
image total	321	83	50	2	58	0	93	607	

Table 4.10: Cohen's κ for family agreement (7 families), with clustered-bootstrap 95% CIs and the Landis & Koch (1977) band. The neutral-labelled subset is degenerate ($\kappa = 0$ by construction) and is omitted.

Subset	n	κ	95% CI	Band
emotional	607	0.23	[0.18, 0.29]	fair
face	402	0.30	[0.23, 0.37]	fair
scene	205	0.13	[0.06, 0.19]	slight
all analysed	1012	0.21	[0.16, 0.25]	fair
distinct emotional	452	0.23	[0.17, 0.29]	fair
distinct face	319	0.29	[0.21, 0.37]	fair
distinct scene	133	0.09	[0.01, 0.17]	slight

Table 4.11: Top-1 family marginals of the two readers over all 1,012 analysed records.

Family	Prompt reader	Image reader
joy	402	461
sadness	99	141
anger	12	89
fear	16	4
surprise	77	108
disgust	1	0
neutral	405	209

4.8 The describer

Table 4.12 crosses the describer's people count with the presence of an expression word over all 1,012 records. The describer answers the expression question for every drawing. It returns an expression word for 406 of the records in which it simultaneously reports that no people are present. Of the 421 scene-classified records, 414 report no people; 406 of those still carry an expression word, and 332 answer "smile", "smiling" or "happy" (80.2% of the no-people records). The smile rate depends on the denominator and is given with both: 57.0% of the 591 face-classified records and 66.1% of all 1,012 analysed records.

The copy rate measures how much the extraction stage adds to the describer's own word. Where the expression answer names an emotion outright (418 records, 41.3% of the corpus), the image reader's top-1 family equals the family that word implies 99.8% of the time. On the broad definition, which admits facial actions such as "smile" and "crying", the copy rate is 68.6%: 82.2% where a person is reported and 48.8% where none is, the "0 people" token pulling the reading towards neutral.

Table 4.12: The describer's answers on all 1,012 analysed records: people count crossed with whether the facial-expression answer contains an expression word. The face rule is the top-left cell. Of the 421 scene-classified records, 414 report no people and 406 of those still contain an expression word (332 answer smile, smiling or happy).

	Expression word present	Expression word absent
People ≥ 1	591	7
People $= 0$	406	8

4.9 Error taxonomy

38 emotional records satisfy the high-confidence criterion, both readers scoring their top family at 0.8 or above and disagreeing; they cover 37 distinct prompts, and since the count fell below the intended sample of 40, every one was examined. Figure 4.6 and Table 4.13 give the primary failure mode assigned from the drawing. By stage, 19 of the 38 trace to the describer (9 where it invented a face or an expression, 10 where it misread one), 5 to the image reader inferring wrongly from a correct description, 12 to the prompt side (11 mixed or ambiguous prompts and 1 prompt-reader error), and 2 to the drawing not depicting the prompt.

Two mechanisms stand out. A people count of zero predicts the invented-expression mode: 13 of the 38 records carry that tag, and seven of the 9 invented-expression records do. And the joy-to-neutral cell is a single mechanism: prompts such as "grateful" and "calm but tired" describe low-arousal positive states for which the seven-family scheme has no home, the describer answers "calm" or "smile", and the reader returns neutral. Visitors also list feelings without connectives ("tired proud happy"), so the automatic tag for mixed prompts,

which looked for “but” and “although”, caught none of the eleven mixed-prompt records; a list detector would serve better.

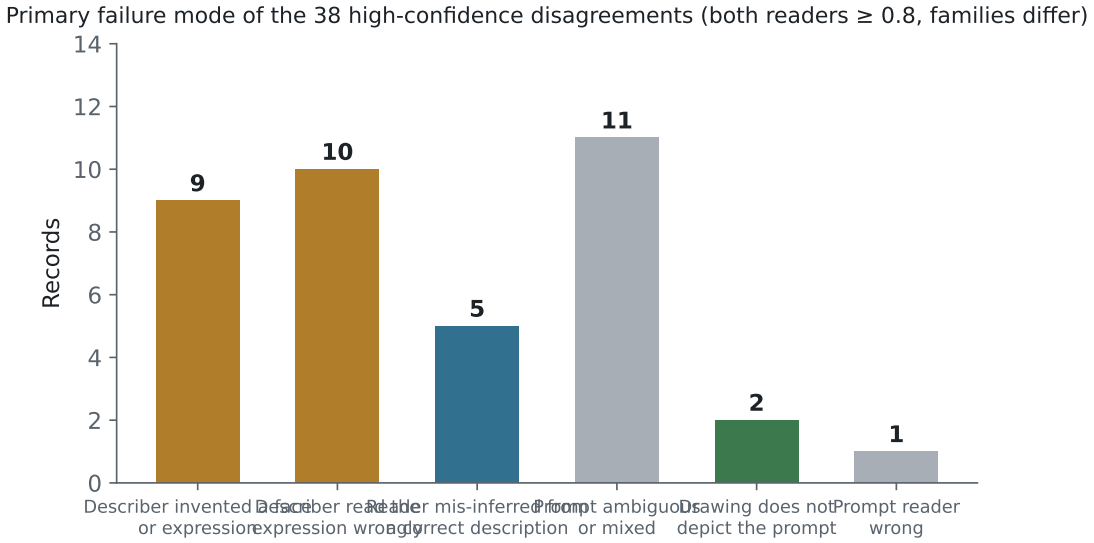


Figure 4.6: Primary failure mode of the 38 high-confidence disagreements, assigned from the drawing.

Table 4.13: Primary failure mode of the 38 high-confidence disagreements (both readers’ top-1 score ≥ 0.8, families differ), assigned from the drawing. Describer faults (a+b) = 19; image reader (c) = 5; prompt side (d+f) = 12; generator (e) = 2.

Mode	Description	n	Exemplar ids
(a)	describer invented a face or expression where none is drawn	9	18, 19, 242
(b)	describer read the expression wrongly	10	39, 192, 202
(c)	reader mis-inferred from a correct description	5	89, 107, 231
(d)	prompt ambiguous or mixed	11	65, 66, 93
(e)	drawing does not depict the prompt	2	137, 771
(f)	prompt reader wrong	1	436

4.10 Description-template ablation

Table 4.14 and Figure 4.7 give the ladder under each altered template, with only the entailment stage re-run. The control re-run of the unmodified template reproduces the baseline exactly, changing the top-1 reading of 0 records, so every difference in the table is a template effect and not numerical noise; the baseline row itself reproduces Section 4.1.

Removing the expression sentence removes nearly all of the agreement. On the emotional prompts family agreement falls from 53.0% to 20.3% (null 16.6%, $z = +3.8$) and valence-sign

agreement from 64.0% to 15.3% (null 11.4%, $z = +5.1$); on the face-classified records valence-sign agreement falls to 18.9%. Fixing the expression word to "unclear" for every record produces the same collapse (family 21.3%, $z = +3.9$), and the reader's output under both variants falls to surprise or neutral for 514 and 368 of the 1,012 records respectively. Removing the action, posture or scene clause moves family agreement by one to three points (50.6%, 51.4%, 53.4%). Removing the people count raises rates and nulls together (scene valence-sign 63.5% against 50.9%, $z = +5.0$): the "0 people" token pulls scene readings towards neutral, and without it the hallucinated smile is read as joy, which most prompts also are.

Under the people-aware template, which describes a drawing as "A drawing showing {scene}." wherever the describer reports no people and leaves every other record unchanged, the face rows are identical to the baseline by construction (73.9%, 61.9%) and the scene rows fall to their nulls: valence-sign agreement 4.7% against 3.4% ($z = +1.6$) and family agreement 18.0% against 17.3% ($z = +0.6$). Restoring the action and posture clauses to those records gives 15.9% against 11.6% ($z = +2.7$) and 21.0% against 16.8% ($z = +2.2$). Over all emotional prompts the people-aware template gives valence-sign agreement of 51.7% ($z = +10.1$) and family agreement of 47.1% ($z = +8.8$).

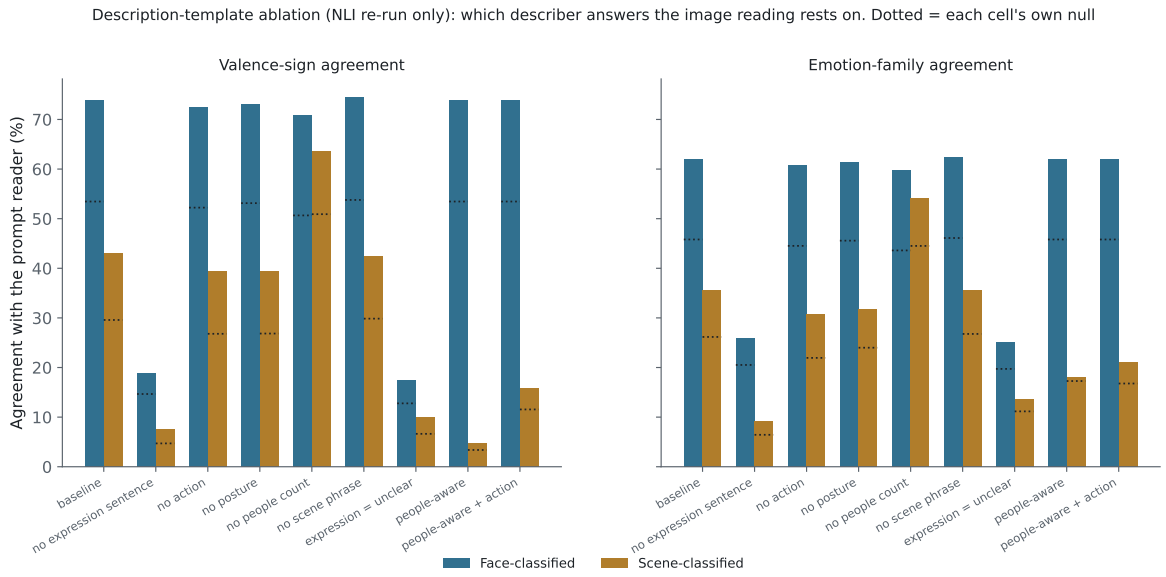


Figure 4.7: Description-template ablation, with only the entailment stage re-run: valence-sign and family agreement by image class under each template, each bar against its own random-pairing null (dotted).

4.11 Single-coder reliability check

The 120-drawing sample has been drawn and laid out on six contact sheets (Figure 3.3); the coding was not complete at the time of writing, and the table in this section is generated from

Table 4.14: Description-template ablation: the NLI stage re-run on descriptions rebuilt from the cached describer answers with one clause removed or fixed; the prompt side and the null procedure unchanged. Valence-sign and family agreement per variant on the emotional prompts and their face/scene halves.

Variant	Subset	n	Sign obs	null	z	Family obs	null	z
baseline	emotional	607	64.0%	45.6%	+11.1	53.0%	38.8%	+10.3
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	42.9%	29.6%	+5.8	35.6%	26.2%	+4.4
drop_expression	emotional	607	15.3%	11.4%	+5.1	20.3%	16.6%	+3.8
	face	402	18.9%	14.7%	+3.8	25.9%	20.5%	+3.8
	scene	205	7.6%	4.7%	+2.7	9.3%	6.4%	+2.2
drop_action	emotional	607	61.9%	43.9%	+11.0	50.6%	36.7%	+10.3
	face	402	72.5%	52.2%	+10.4	60.7%	44.5%	+10.4
	scene	205	39.4%	26.8%	+5.6	30.7%	21.9%	+4.6
drop_posture	emotional	607	62.3%	44.5%	+10.7	51.4%	37.9%	+9.7
	face	402	73.1%	53.1%	+10.0	61.4%	45.6%	+10.1
	scene	205	39.4%	26.8%	+5.6	31.7%	24.0%	+3.6
drop_people	emotional	607	68.5%	50.7%	+10.8	57.8%	43.7%	+9.9
	face	402	70.8%	50.7%	+10.4	59.7%	43.6%	+10.3
	scene	205	63.5%	50.9%	+5.0	54.1%	44.5%	+3.8
drop_scene	emotional	607	64.2%	46.0%	+10.7	53.4%	39.2%	+9.8
	face	402	74.4%	53.8%	+10.4	62.4%	46.1%	+10.1
	scene	205	42.4%	29.9%	+5.6	35.6%	26.8%	+3.9
fixed_expression	emotional	607	15.1%	10.8%	+5.9	21.3%	17.5%	+3.9
	face	402	17.5%	12.8%	+4.4	25.1%	19.7%	+4.0
	scene	205	10.0%	6.6%	+2.8	13.7%	11.2%	+1.5
people_aware	emotional	607	51.7%	36.9%	+10.1	47.1%	34.5%	+8.8
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	4.7%	3.4%	+1.6	18.0%	17.3%	+0.6
people_aware_no_expression	emotional	607	55.3%	39.5%	+10.5	48.1%	35.0%	+9.6
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	15.9%	11.6%	+2.7	21.0%	16.8%	+2.2
control_full_template	emotional	607	64.0%	45.6%	+11.1	53.0%	38.8%	+10.3
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	42.9%	29.6%	+5.8	35.6%	26.2%	+4.4

the scored file once it is. The table reports the describer's people count against the coder's, the face rule's precision and recall against the coder's face judgement, what lies behind the describer's "smiling" answers, the image reader's valence sign against the coder's, and the face/scene contrast recomputed on the coder's split rather than the describer's, which is the one estimate in the dissertation that breaks the circularity of Section 3.4 for the records it covers. It is a single coder, and it is stated as such.
[Human-check table pending: code `outputs/human_check/labels.csv`*, run* `human_check.py score`*, then* `export_tex.py`*.]*

4.12 Summary of findings

1. On the 607 prompts the text reader labels emotional, the two readers agree far above chance on every rung of the ladder: valence sign 64.0% against 45.6%, family 53.0% against 38.8%; the result survives clustering, the distinct-prompt re-run and the Holm correction (objectives 2 and 3).
2. Where the describer reports a readable face the agreement is far higher than where it reports a scene, by +30.9 pp on valence sign (CI [+22.6, +40.6], $p = 0.0001$); the scene result weakens on distinct prompts and, under a people-aware description, falls to its null (objectives 3, 4 and 7).
3. Prompts the text reader labels neutral sit at chance on the one measure that can move there, 76.6% against 76.5%, and remain there when the 68 prompts that contain an affect word are removed (objective 3).
4. Like for like, describe-then-extract agrees with the prompt reader on 53.0% of families against 22.4% for direct CLIP, a difference of +30.6 pp with $p < 0.001$ on the clustered test (objective 5).
5. When the same words are drawn again, the image reader returns the same family for 68.8% of drawings in a group against 59.4% by chance ($z = +4.5$): more consistent than chance and far from fixed (objective 6).

The audits qualify all five. The describer names an expression for 406 of the 414 no-people scene records, the image reader copies an emotion-naming expression word 99.8% of the time, and the ablation shows that the reading rests on that word: the split and most of the agreement run through it.

Chapter 5

Discussion

5.1 What the results establish

Three things, and I want to be careful not to claim a fourth.

First, the emotion a text model recovers from a visitor's words and the emotion a second model recovers from the drawing made of those words agree far more often than chance would produce, on every rung of the ladder, and the effect is not an artefact of the things that most often produce artefacts in a corpus like this. It survives clustering on the prompt group, which takes the 249 repeated records out of the count of independent evidence; it survives re-running on the first occurrence of every prompt; and it survives a Holm correction across every non-degenerate comparison the analysis prints. The valence-sign result, 64.0% against 45.6%, is the one I would defend in any room, because it asks the coarsest question and because its denominator is stated.

Second, the agreement is concentrated in drawings the describer reads as containing a face. Where it reports a scene, the agreement in the main table is weak, it weakens further on distinct prompts, and its family cell does not survive the multiplicity correction. The ablation then removes it: describe those drawings without the person and expression the template inserted, and this reader finds nothing above chance in them. That is a stronger statement than the one the project set out to make, and a different one. The design expected faces to agree more than scenes. What it found is that, for this instrument, faces agree and scenes do not.

Third, the instrument behaves as an instrument should on its checks. Prompts the text reader could not read affect from sit at chance on the one measure with a live null, and stay there when the prompts that plainly contain an affect word are removed. And, scored like for like, the describe-then-extract reader agrees with the prompt reader far more often than a direct CLIP reader does, by +30.6 pp on family with the clustered test at $p < 0.001$. That last result has to be framed with care. Neither image reader has ground truth. What the comparison shows is that two ways of reading a drawing disagree with each other on most records, and that one of them sits much closer to a third model's reading of the words. It does not show that either reader is more correct, and I have avoided the word throughout.

The fourth thing, the one not claimed, is any statement about what a person sees. Every number in this dissertation is the agreement of two machine readers. The human study that would turn agreement into perception is designed, its sample is drawn, and it is not done.

5.2 The describer is the weak link

The audits were written to find out where the instrument bends, and they all point at the same place. The description template asserts a person and a facial expression for every drawing. The describer obliges: for 414 scene-classified drawings it reports that nobody is present and, in 406 of them, names an expression anyway, “smile”, “smiling” or “happy” in 332 cases. The image reader then reads that sentence. Where the expression word names an emotion outright, which it does for 41.3% of the corpus, the reader’s top family is the family that word implies 99.8% of the time. The two-stage architecture was meant to keep the vision model away from the emotion judgement; the copy rate says that, for those records, the second stage relays the first.

The ablation says it more directly. Take the expression sentence out and family agreement on the emotional prompts falls from 53.0% to 20.3%; fix the expression word to “unclear” and it falls to 21.3%; take out any other clause and it moves by a point or three. The reading rests on one word. In effect the pipeline asks the describer what the face is doing and maps the answer to a family, with the entailment model as an expensive lookup table. That is a finding about the instrument rather than a failure of the project, and I would rather report it than have it found by an examiner, but it changes how the method comparison should be read: the advantage over direct CLIP may lie in BLIP-VQA being a better reader of drawn faces than CLIP is, not in the describe-then-extract architecture as such. An ablation that stripped emotion words from the expression answer before extraction would separate the two, and is listed as further work.

The error taxonomy closes the loop from the other direction. Of the 38 disagreements on which both readers were confident, 19 trace to the describer, 5 to the reader, 12 to the prompt side and only 2 to the drawing failing to depict the prompt. The generator, in other words, is rarely the reason the two readings differ; the readers are. This sits well with what the perception benchmarks report about vision–language models: a systematic bias towards positive labels [6], which appears here as a smile rate of 57.0% on face-classified records and 66.1% corpus-wide, and a confusion of what a subject expresses with what a viewer feels [24], which the describer’s invented expressions are an extreme form of.

5.3 Circularity of the face/scene split

The face/scene split was my supervisor’s hypothesis and the design’s main contrast, and it needs its own paragraph of honesty. The split is defined from the describer’s output: a person reported, and an expression word present. It is driven almost entirely by the people count, since 414 of the 421 scene-classified records are ones where the describer said nobody was

there. The image reader then reads the same description. The contrast between the two classes is therefore a statement about the describer's output, not about faces in drawings, and it is entangled with its own definition. It is real in the sense that matters statistically: it clears its nulls, it survives clustering, and the permutation test puts it at $p = 0.0001$. But what it establishes is that when the describer reports a face and an expression, the reader's family tracks the prompt, and when the describer reports no person, the reader has only an invented expression to go on and, once that is removed, nothing. Whether the scene drawings hold a feeling that a person can read is a question this design cannot answer, and the single-coder check, which classifies drawings without reference to the describer, is the only estimate in the project that can. Until it is scored, the right form of words is the one used in Chapter 4: faces, and only faces, for this reader.

5.4 What is not claimed

Four non-claims, each with its reason.

No direction between positive and negative feelings. The first build reported that negative feelings survived the drawing and positive ones collapsed; the second build, with a different image reader, reported the reverse. Both were wrong to report anything: within a single-valence subset the valence-sign null is degenerate, so there is nothing to test, and the raw difference on the loosest measure is a base-rate effect of where the families sit on the plane. Two readers that point in opposite directions cannot settle the question between them.

No statement about human perception. The word in the registered title is a question, not a result; the dissertation measures agreement between readers and says so on every page it can.

No claim about colour. The corpus is monochrome throughout and there is no colour condition. The colour literature [2] explains why a monochrome corpus is a hard case, and that study models colour features only and is silent on line; nothing here can be attributed to the absence of colour rather than to the readers, the taxonomy or class rarity.

No claim about the generator alone from the consistency result. The 68.8% within-group agreement against 59.4% bounds the whole chain from prompt to reading. With no reliability estimate for the image reader, an unknown share of the variation is measurement noise, and with generation parameters unrecorded, the rest cannot be attributed to sampling rather than to prompt rewriting.

There is also a restriction rather than a non-claim. Fear (16 prompts), anger (12) and disgust (1) are too rare on the prompt side to support a family-level statement, the image reader never returns disgust and returns fear 4 times, and the surprise family is largely curiosity-mapped prompts under the standard label grouping. Claims are confined to overall agreement and the face/scene contrast.

5.5 Relation to prior work

The closest study is Lomas and colleagues' human rating of emotional alignment for three generators [49]. Their strongest secondary finding was that emotions rendered on a person aligned better than the same emotions rendered on a robot. The present project finds the same direction with machine readers and unmediated prompts, and then, through the ablation, finds that its own version of the effect is carried by a face being reported rather than by a face being drawn. The two results are consistent and neither validates the other, because theirs used human raters and templated prompts and this used machine readers and real ones; the human study that would connect them is the one the pipeline was built to precede.

The reasoning behind the ladder comes from Cowen and Keltner [19]: if experience is a set of categories bridged by gradients, near misses carry information. The results support the choice. Valence–arousal similarity and top-3 overlap both sit far above their nulls where family agreement is only 53.0%, which says that the readers are close far more often than they coincide, and a right-or-wrong score would have called that closeness error.

The prompt reader's weakness on this register is the bound on everything downstream, exactly as the text literature predicts [21, 84]. A kappa of 0.23 on the emotional prompts, in the fair band, is the kind of figure reported between human annotators for fine-grained emotion labelling of text, where GoEmotions' own raters agreed only moderately on many categories; two automatic readers separated by a generator and a drawing were never going to do better than the annotators of the training data did with each other. Carrasco and colleagues [10] report that human agreement with a generator's emotion labels rises sharply once emotions are collapsed to positive and negative; the same pattern appears here in the gap between the valence-sign and family rungs.

The decision to read every value against a null follows Zhang and Wan [90] and the holistic evaluation of Lee and colleagues [41], who show that automatic image–text scores move under trivial perturbations and correlate weakly with people. A score that changes its verdict under a one-pixel shift should not be quoted bare, and neither, this project argues, should a similarity whose floor is 0.41. Direct CLIP is the right baseline for the same reason CLIPScore [29] is the field's default, and it fails here for the reason VQAScore's authors give for preferring a question-answering judge [47]: an embedding trained on captioned photographs has no purchase on a monochrome drawn face. The describer is, in that sense, a VQAScore-style judge pointed at expression rather than at objects, which is also why its expression answer does the work.

Finally, the positive bias the benchmarks find in vision–language models [6] is reproduced here as a smile rate and as invented expressions, and WikiArt Emotions' observation that images with faces yield more consistent emotion judgements [59] is reproduced as the face/scene contrast, with the caveat that the consistency here is a machine's.

5.6 Limitations

1. There is no human anchor except the single-coder check, and that check is a single coder. Every result is reader-to-reader agreement.
2. The emotional/neutral split is the prompt reader's judgement, never human-checked; 16.8% of the control contains an affect word, so the control is "prompts the reader could not read affect from" rather than "prompts with no feeling".
3. The face/scene split is circular: defined from the description the image reader reads, and driven by the people count.
4. The description template asserts a person and an expression for every drawing. Its consequence was measured, not designed away; a people-aware template is the obvious default for any future run.
5. Seven families have no home for low-arousal positive states, so "calm", "grateful" and "at peace" fall to neutral.
6. The family centroids are hand-placed; per-family means from the NRC norms [56] would replace a judgement with data.
7. The z-scores come from 300 re-pairings and are ordering statistics; across seeds the headline moves between +11.2 and +12.2.
8. Generation parameters are unknown, so nothing can be regenerated and the consistency result cannot be decomposed.
9. Redaction leaves the drawing and the residual text in place, so re-identification risk is reduced, not removed.
10. One machine and one run: there is no second describer and no test–retest estimate for either reader.

5.7 Threats to validity

Construct validity is the governing threat: agreement between two readers is not affect as perceived, and the dissertation does not pretend otherwise. Internal validity is threatened by the leak from describer to reader and by the circular split, both of which are now quantified rather than assumed, and by pseudo-replication from repeated prompts, which the clustered inference addresses. External validity is limited to one generator, one festival, one language for the most part, and monochrome output; nothing here transfers to colour or to another model without being re-measured. Statistical validity is threatened by the number of comparisons, addressed by the Holm procedure, and by degenerate nulls, now detected from the null's own spread. The threats the design addressed are stated with the numbers that address them; the ones it could not address, above all the human anchor, are stated as the first items of further work.

Chapter 6

Conclusions and Further Work

6.1 Conclusions

The question was whether the emotion recoverable from a line drawing agrees with the emotion recoverable from the words that produced it. The answer, bounded exactly as Chapter 5 bounds it, is that two automatic readers agree far more often than chance on the 607 prompts that carry a readable feeling, that the agreement is carried by drawings in which the describer can read a face, and that for this instrument scenes contribute nothing above chance once an invented expression is removed from their description.

Against the objectives of Section 1.3: a describe-then-extract image reader was built and run over all 1,025 images (objective 1). A four-rung similarity ladder was defined and every rung given a random-pairing null, which turned out to be the most consequential decision in the project, since the loosest rung sits at 76.6% for random pairs (objective 2). Agreement was measured on the whole corpus and on both splits: 64.0% valence-sign agreement against 45.6% on emotional prompts, 73.9% on face-classified drawings against 42.9% on scene-classified, and chance on the neutral-labelled control (objective 3). The face/scene difference was tested at +30.9 pp with $p = 0.0001$, and the headline results survive the distinct-prompt re-run and a Holm correction while the scene family result does not (objective 4). Like for like, describe-then-extract agrees with the prompt reader on 53.0% of families against 22.4% for direct CLIP (objective 5). The repeated prompts gave a consistency of 68.8% against a null of 59.4%, more consistent than chance and far from fixed (objective 6). The audits found that the describer names an expression for drawings with no person, that the reader copies an emotion-naming expression word 99.8% of the time, and that the reading rests on that one word; the error taxonomy placed most confident disagreements at the describer and almost none at the generator; the single-coder sample is drawn and awaits coding (objective 7). The toolkit, its two-layer tests and the demonstrator are released (objective 8).

What the dissertation contributes is not a metric. It is a corpus and a question that nobody had put together, a way of asking the question that gives every value its chance level, a consistency experiment the repeated prompts made possible, and an audited account of an instrument that ends up more modest than its architecture promised. The most useful

sentence in it may be the one the ablation forced: the two-stage reader is, for most records, a facial-expression question with a lookup table attached, and a study that wants to know what a drawing shows will need either a person or a describer that can read a scene.

6.2 Further work

In priority order, each tied to a result above.

1. **The human-rated study.** The pipeline was built to precede it. Raters would score the same ladder on a stratified sample of drawings, blind to provenance given the labelling effects reported by Ragot, Horton and Millet [70, 32, 54], so that machine and human agreement can be compared rung by rung; Lomas and colleagues' protocol [49] is the design to adapt, and the single-coder check is its pilot.
2. **A people-aware description template as the default.** The ablation is its evidence. It should replace the current template in the pipeline and the demonstrator together, so the two stay in parity, and the main table should be re-run under it.
3. **An expression-word ablation on the face-classified records.** Strip the emotion word from the expression answer before extraction and measure what remains; this is the experiment that separates "BLIP-VQA reads drawn faces well" from "describe-then-extract is a better architecture".
4. **A scene-capable describer.** A sketch- or drawing-specific model was designed and not executed for want of time; if the human study finds that scene drawings do hold a readable feeling, a describer that reads affect from composition, posture and setting rather than from a face is the main route to measuring it, and it would feed the people-aware template.
5. **A second describer and a test–retest estimate for both readers,** so that the consistency result can be decomposed into generator variance and measurement noise.
6. **NRC per-family valence–arousal means** in place of the hand-set centroids [56], and a richer scheme, or the 27 categories of Cowen and Keltner [19], so that low-arousal positive states no longer fall to neutral.
7. **A second coder** for the reliability check, with an agreement statistic, so that the human anchor is itself measured.

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Appendices

Appendix A

Reproducibility

A.1 Repository layout

- `emoviz/emoviz/metrics.py`: the single definition of every measure, split rule, centroid and the random-pairing null; imported by every script.
- `emoviz/emoviz/data.py`: the loader: filename schemes, timestamps, prompt groups, name redaction at load time.
- `emoviz/preprocess_names.py`: the three-stage redaction procedure; the decision table travels with the dataset.
- `emoviz/describe_extract.py`: the pipeline (BLIP-VQA → template → BART-MNLI); crash-safe, resumable.
- `emoviz/analysis_final.py`: splits, ladder, nulls, distinct-prompt re-run; writes `outputs/analysis_final.json`.
- `emoviz/significance.py`: clustered bootstrap and clustered permutation test of the face/scene difference.
- `emoviz/inference.py`: clustered CIs for every subset, Cohen's κ, Holm–Bonferroni, seed stability.
- `emoviz/method_comparison.py`: like-for-like image-reader comparison with paired tests.
- `emoviz/data_audit_consistency.py`: data accounting and the repeated-prompt consistency with its permutation null.
- `emoviz/audits.py`: control contamination, describer leak and copy rate, per-family recall.
- `emoviz/error_taxonomy.py`: the high-confidence disagreement census and failure modes.
- `emoviz/ablate_nli.py`: description-template ablation (NLI stage only).

- `emoviz/human_check.py`: the single-coder sample, contact sheets and scoring.
- `emoviz/make_figures.py` and `emoviz/export_tex.py`: every figure and every number in this document, generated from the JSON result files.
- `emoviz/app.py` and `emoviz/web/`: the demonstrator.
- `emoviz/test.sh` and `emoviz/test_numeric.py`: the smoke suite and the pinned numeric checks.
- `emoviz/outputs/`: cached model outputs, result JSON, figures, the frozen direct-CLIP baseline.

A.2 Run order

1. `bash emoviz/setup.sh`: creates the environment from the pinned `requirements.txt`.
2. `describe_extract.py`: the pipeline over all images (cached; re-run only to regenerate).
3. `analysis_final.py`: the main analysis (seconds; uses the cached outputs).
4. `significance.py`, `inference.py`, `method_comparison.py`
5. `data_audit_consistency.py`, `audits.py`, `error_taxonomy.py`, `ablate_nli.py`
6. `human_check.py sample`; code `outputs/human_check/labels.csv`; `human_check.py score`
7. `make_figures.py`; `export_tex.py`; then build this document (`pdflatex`, `bibtex`, `pdflatex` twice).
8. `bash emoviz/test.sh`: runs everything above that is fast, plus `test_numeric.py`.

All scripts are run with `emoviz/.venv/bin/python` from the `emoviz/` directory.

A.3 Environment

Appendix B

Instruments

B.1 The five describer questions

The describer is asked, in this order, and its answers are stored per record:

1. *What is the facial expression?* (`expression`)
2. *What is the person doing?* (`action`)
3. *What is the body posture?* (`posture`)
4. *How many people are there?* (`people`)
5. *What is in the picture?* (`scene`)

The word "emotion" appears in none of them.

B.2 The description template

The five answers are composed into one sentence:

```
A drawing of {people} showing {scene}.  The person is {action}, {posture}.
The facial expression is {expression}.
```

The template is applied to every record, including those where `people` is "0" or "none"; the ablation variants remove one clause at a time or fix `expression` to "unclear".

B.3 The image reader's hypothesis

Zero-shot classification by entailment with the hypothesis template

```
This scene conveys {family}.
```

scored for each of the seven families; the three highest entailment scores are kept.

B.4 GoEmotions labels to families

joy admiration, amusement, approval, caring, desire, excitement, gratitude, joy, love, optimism, pride, relief

sadness disappointment, embarrassment, grief, remorse, sadness

anger anger, annoyance, disapproval

fear fear, nervousness

surprise surprise, realization, confusion, curiosity

disgust disgust

neutral neutral

Family scores are the sums of the member probabilities, normalised to a distribution.

B.5 Family centroids on the circumplex

Family	Valence	Arousal
joy	+0.80	0.60
sadness	−0.75	0.30
anger	−0.70	0.80
fear	−0.65	0.75
surprise	+0.30	0.80
disgust	−0.70	0.60
neutral	0.00	0.30

Hand-set from the standard placement of the basic emotions on the circumplex; the minimum of Eq. 3.1 over these points is 0.412.

B.6 Readable expression words

A record is face-classified when the describer reports at least one person and its expression answer contains one of:

> smile, smiling, happy, sad, angry, serious, surprised, surprise, calm, excited, crying, laughing, frowning, scared, shocked, worried, concentration, concentrating, grin, grinning, neutral, confused, determined, pensive, sleepy, sleeping.

The people count is parsed from digits or number words (none, one, two, . . . , several, many, crowd, group, lots); a non-empty answer that is neither is read as one person present.

B.7 Name redaction procedure

1. **Detect.** Candidates from three detectors: a given-name gazetteer (7,944 names), mid-sentence capitalised tokens absent from a 236k-word English dictionary, and naming phrases ("called X", "my friend X"). High recall by design, because most prompts are written entirely in lower case.
2. **Review.** Every candidate has a recorded decision (redact or keep) with a reason and, for redactions, the record it applies to. The audit fails if any candidate is unreviewed or if a redaction entry matches nothing.
3. **Apply.** At load time, per record, replacing the token with `[name]`; consecutive placeholders collapse to one. The files on disk are never modified. Illness, grief and place names are kept.

B.8 Single-coder check: coding instructions

The coder works from numbered contact sheets that show only the drawings, and fills four columns per drawing:

person y / n — is at least one person or face drawn?

face y / n — is there a face whose expression can be read? (n if no person)

valence + / − / 0 — the feeling the drawing gives: positive, negative, neither

family joy / sadness / anger / fear / surprise / disgust / neutral

The sample is fixed by seed (120 drawings: 40 face-classified, 40 scene-classified with no people reported, and every record with a vague or word-only people answer, topped up equally from the first two strata). The prompt, the description and the model readings are not shown during coding.